

AOS-CX 10.06 VXLAN EVPN Guide

6200, 6300, 6400, 8325, 8360, 8400 Switch Series



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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A) ***Supports static VXLANs only.***
- Aruba 6300 Switch Series (JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, JL668A, JL762A)
- Aruba 6400 Switch Series (JL741A, R0X26A, R0X27A, R0X29A, R0X30A)
- Aruba 8325 Switch Series (JL624A, JL625A, JL626A, JL627A)
- Aruba 8360 Switch Series (JL700A, JL701A, JL702A, JL703A, JL706A, JL707A, JL708A, JL709A, JL710A, JL711A)
- Aruba 8400 Switch Series (JL375A, JL376A) ***Does not support static VXLANs.***

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in **Support and other resources**.

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
<code>example-text</code>	In code and screen examples, indicates text entered by a user.
Any of the following:	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code:
• <code><example-text></code>	• For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
• <code><example-text></code>	
• <i>example-text</i>	
• <i>example-text</i>	

Table Continued

Convention	Usage
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or	Ellipsis:
...	- In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. - In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment. The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where `<VLAN-ID>` is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

member/slot/port

On the 6200 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Line module number. Always 1.
- *port*: Physical number of a port on a line module.

For example, the logical interface 1/1/4 in software is associated with physical port 4 in slot 1 on member 1.

On the 6300 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 10. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Line module number. Always 1.
- *port*: Physical number of a port on a line module.

For example, the logical interface 1/1/4 in software is associated with physical port 4 in slot 1 on member 1.

On the 6400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/1 and 1/2.
 - Line modules are on the front of the switch starting in slot 1/3.
- *port*: Physical number of a port on a line module.

For example, the logical interface 1/3/4 in software is associated with physical port 4 in slot 3 on member 1.

On the 83xx Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Line module number. Always 1.
- *port*: Physical number of a port on a line module

For example, the logical interface 1/1/4 in software is associated with physical port 4 in slot 1 on member 1.



NOTE: If using breakout cables, the port designation changes to x:y, where x is the physical port and y is the lane when split to 4 x 10G or 4 x 25G. For example, the logical interface 1/1/4:2 in software is associated with lane 2 on physical port 4 in slot 1 on member 1.

On the 8400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/5 and 1/6.
 - Line modules are on the front of the switch in slots 1/1 through 1/4, and 1/7 through 1/10.
- *port*: Physical number of a port on a line module

For example, the logical interface 1/1/4 in software is associated with physical port 4 in slot 1 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: *member/power supply*:
 - *member*: 1.
 - *power supply*: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: *member/tray/fan*:
 - *member*: 1.
 - *tray*: 1 to 4.
 - *fan*: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: *member/module*:
 - *member*: 1.
 - *member*: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Virtual eXtensible LAN (VXLAN) is a MAC-in-UDP technology that provides layer 2 connectivity between networks across an IP network. VXLAN is typically used to extend layer 2 segments across an entire data center or between remote data centers. It is also used to provide multi-tenant services, where the same IP/MAC addresses or VLANs are used on different network segments.

VXLAN benefits

- Supports more virtual switched domains than VLANs: Each VXLAN is uniquely identified by a 24-bit VXLAN ID, called the VXLAN Network Identifier (VNI). The total number of VXLANs can reach 16 million. This makes VXLAN a better choice than 802.1Q VLAN for isolating traffic for user terminals.
- A VXLAN segment differentiates individual logical networks so that several isolated layer 2 VXLAN networks can coexist on a common layer 3 infrastructure. (Hosts in different VXLAN segments cannot communicate with each other. Each VXLAN segment can be treated as a tenant in a data center topology.)
- Easy to deploy and maintain: VXLAN only needs to be deployed on the edge devices of the transport network. Devices in the transport network perform typical layer 3 forwarding.

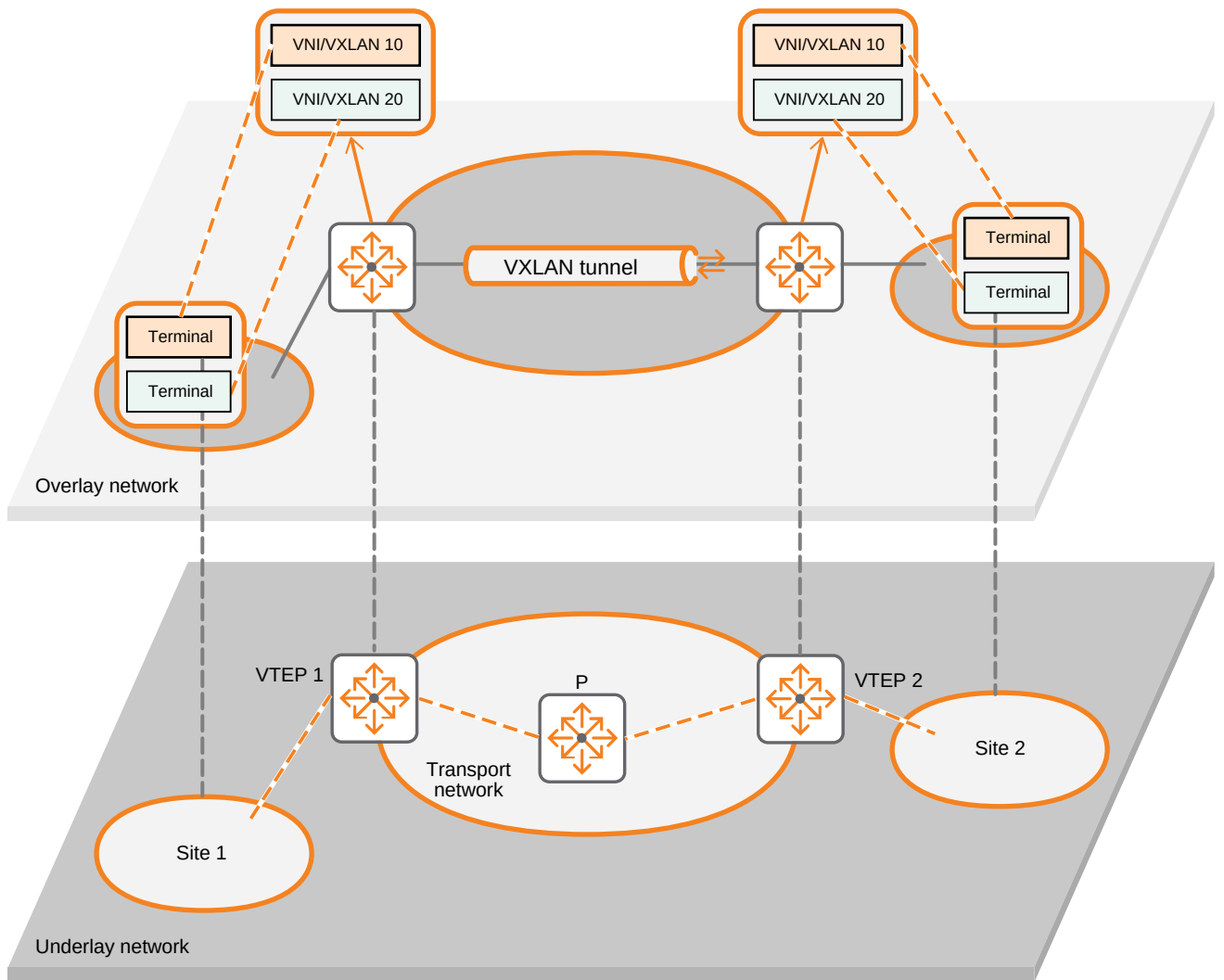
VXLAN network model

A VXLAN is a virtual layer 2 network (known as the overlay network) built on top of an existing physical layer 3 network (known as the underlay network). VXLAN uses a logical tunnel to transport traffic between two endpoints.

VXLAN encapsulates layer 2 Ethernet frames in layer 3 UDP packets, meaning virtual layer 2 subnets can span underlying layer 3 networks. A VXLAN network identifier (VNI) is used to segment each layer 2 subnet in a similar way as traditional VLAN IDs.

A VXLAN tunnel endpoint (VTEP) is a VXLAN-capable device that encapsulates and de-encapsulates packets. In the physical network, a switch typically functions as a layer 2 or layer 3 VXLAN gateway and is considered a hardware VTEP. The virtual network equivalents are known as software VTEPs, which are hosted in hypervisors such as VMware ESXi or vSphere.

A tunnel endpoint is called a virtual tunnel endpoint (VTEP). VTEPs are responsible for encapsulation and decapsulation of network traffic.



Static VXLAN uses flood and learn (or ingress replication) to learn the MACs of the remote host, which involves manual configurations of remote VXLAN Tunnel End Points (VTEPs) in the flood list. MP-BGP EVPN is used to discover remote VTEPs and advertise MAC address and MAC/IP bindings in the VXLAN overlay, thus eliminating the flood and learn. MP-BGP supports a new EVPN Network Layer Reachability Information (NLRI) carried in BGP using Multiprotocol BGP Extensions with a newly defined Address Family Identifier (AFI) and Subsequent Address Family Identifier (SAFI). Route Distinguisher (RD) is a unique number prepended to the advertised address within the VRF, ensuring support for overlapping IP addresses and MACs across different tenants. Routes can be imported and exported across VLANs and VRFs using a BGP extended community called Route Target (RT) that are advertised along with the EVPN routes.



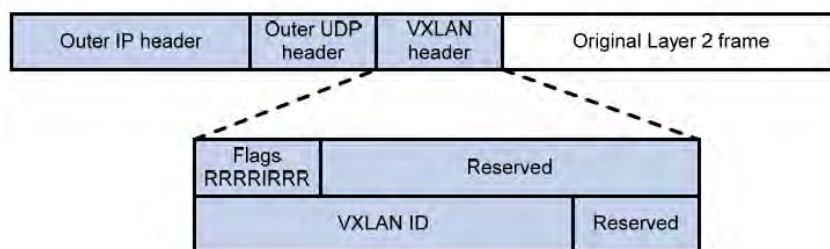
NOTE:

- Configuring static VTEPs is not supported when EVPN is enabled.
- The 6200 Switch Series supports static VXLAN only.
- VMware NSX-V integration and EVPN are mutually exclusive and cannot be configured together.
- EVPN control pane is supported on the 6300, 6400, 8325, 8360, and 8400 Switch Series.

VXLAN packet format

A VTEP encapsulates a frame in the following headers:

- 8-byte VXLAN header: VXLAN information for the frame.
 - 24-bit VXLAN Network Identifier: Identifies the VXLAN of the frame. It is also called the virtual network identifier (VNI).
 - 8-bit flags: If the fifth bit (I flag) is 1, the VNI is valid. If the I flag is 0, the VNI is invalid. All other bits are reserved and set to 0.
- 8-byte outer UDP header for VXLAN: The default VXLAN destination UDP port number is 4789.
- 20-byte outer IP header: Valid addresses of VTEPs or VXLAN multicast groups on the transport network. Devices in the transport network forward VXLAN packets based on the outer IP header.
- 24-bit and 8-bit reserved fields.



Supported RFCs

VXLAN

- RFC 7348 - Virtual eXtensible Local Area Network (VXLAN)
- RFC 8365 - EVPN over NVE (VXLAN)

EVPN

- RFC 4760 - Multiprotocol Extensions for BGP-4
- RFC 7432 - BGP MPLS-Based Ethernet VPN

Static VXLAN (also known as unicast VXLAN), is the easiest way to connect two VTEPs. In this method, the VXLAN uses a flooding-and-learning technique in the VXLAN data plane to learn the address of hosts.

Static VxLAN for IPv6 overlay is not supported. IPv4 overlay over static VxLAN has limitations with silent hosts.

Configuring a static VXLAN

Prerequisites

- A enabled layer 3 interface with an IP address assigned to it, created with the command `interface`.
- A layer 3 VLAN.

Procedure

1. Create loopback interface 1 using the command `interface loopback` and assign a source IP to it.
2. Create VXLAN interface 1 with the command `interface vxlan`.
3. Enable VXLAN interface 1 with the command `no shutdown`.
4. Set the source IP address for the VXLAN tunnel with the command `source ip`.
5. Create a VNI with the command `vni`.
6. Define the IP address of the peer VTEP with the command `vtep-peer`.
7. Review VXLAN settings with the commands `show interface vxlan1`, `show interface vxlan vni`, and `show interface vxlan vteps`.

Example

This example creates a static VXLAN as follows:

- Creates loopback interface 1 with IP address 10.1.1.1/24.
- Creates VLAN 2.
- Creates VXLAN interface 1.
- Sets the source IP address for the VXLAN tunnel to 10.1.1.1.
- Creates VNI 2.
- Sets the VTEP peer to 200.1.1.1.
- Associates VLAN 2 with VNI 2.

```
switch(config)# loopback 1
switch(config-if)# ip address 10.1.1.1/24
switch(config-if)# exit
```

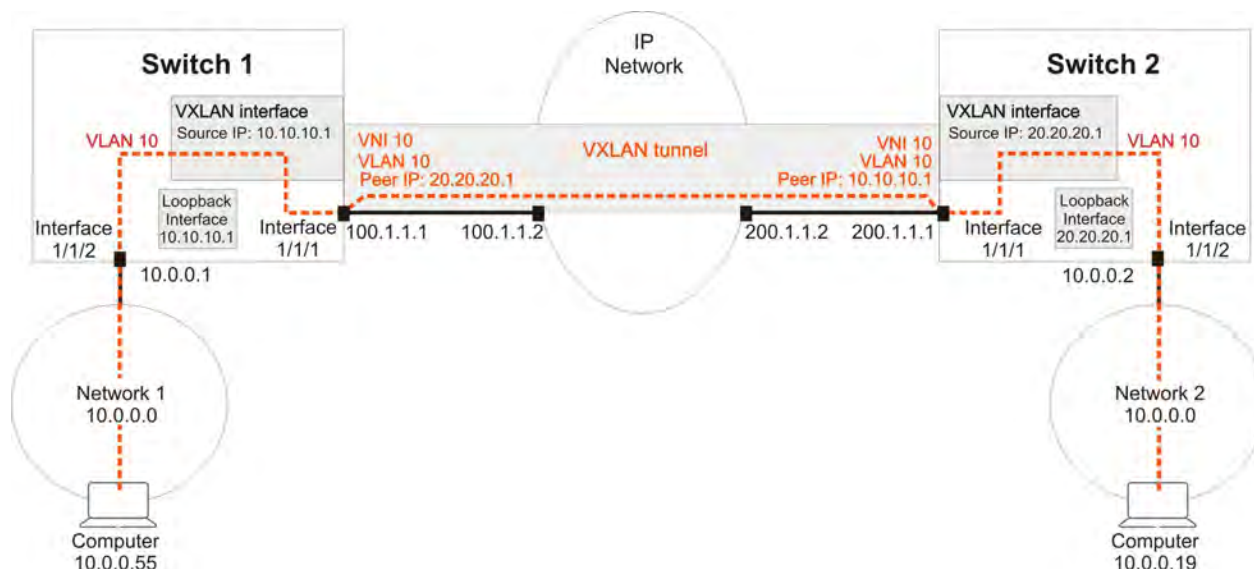
```

switch(config)# interface vxlan 1
switch(config-vxlan-if)# no shutdown
switch(config-vxlan-if)# source ip 10.1.1.1
switch(config-vxlan-if)# vni 2
switch(config-vni-2)# vtep-peer 200.1.1.1
switch(config-vni-2)# vlan 2
switch(config-vni-2)# exit
switch(config-vxlan-if)# exit
switch(config-if)# exit

```

Static VXLAN with a single VNI between two VTEPs

This example creates a static VXLAN tunnel between two switches, enabling traffic from two networks to traverse an underlay IP network.



Procedure

1. On switch 1:

- Enable loopback interface 1 and assign the IP address 10.10.10.1/24 to it.

```

switch# config
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 10.10.10.1/24
switch(config-if)# exit
switch(config)#

```

- Enable interface 1/1/1 and assign the IP address 100.1.1.1/24 to it.

```

switch(config)# interface 1/1/1
switch(config-if)# routing (6300/6400 only)
switch(config-if)# ip address 100.1.1.1/24
switch(config-if)# no shutdown
switch(config-if)# exit
switch(config)#

```

- Create VLAN 10.

```

switch(config)# vlan 10
switch(config-vlan-10)# exit

```

- d. Enable interface 1/1/2 and assign VLAN 10 to it.

```
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# no routing (8325/8360/8400 only)
switch(config-if)# vlan access 10
switch(config-if)# exit
```

- e. Create VXLAN interface 1 and assign the loopback source IP address 10.10.10.1 to it.

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# source ip 10.10.10.1
switch(config-vxlan-if)# no shutdown
```

- f. Create VNI 10, and assign the VTEP peer address 20.20.20.1 and VLAN 10 to it.

```
switch(config-vxlan-if)# vni 10
switch(config-vni)# vtep-peer 20.20.20.1
switch(config-vni)# vlan 10
switch(config-vni)# exit
switch(config-vxlan-if)# exit
```

- g. Add a route to the peer VTEP peer address 20.20.20.1.

```
switch(config)# ip route 20.20.20.1 100.1.1.2
```

2. On switch 2:

- a. Enable loopback interface 2 and assign the IP address 20.20.20.1/24 to it.

```
switch# config
switch(config)# interface loopback 2
switch(config-loopback-if)# ip address 20.20.20.1/24
switch(config-if)# exit
switch(config)#
```

- b. Enable interface 1/1/1 and assign the IP address 200.1.1.1/24 to it.

```
switch# config
switch(config)# interface 1/1/1
switch(config-if)# routing (6300/6400 only)
switch(config-if)# ip address 200.1.1.1/24
switch(config-if)# no shutdown
switch(config-if)# exit
```

- c. Create VLAN 10.

```
switch(config)# vlan 10
switch(config-vlan-10)# exit
```

- d. Enable interface 1/1/2 and assign and VLAN 10 to it.

```
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# no routing (8325/8360/8400 only)
switch(config-if)# vlan access 10
switch(config-if)# exit
```

- e. Create VXLAN interface 1 and assign the source IP address 20.20.20.1 to it.

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# source ip 20.20.20.1
switch(config-vxlan-if)# no shutdown
```

- f. Create VNI 10, and assign the VTEP peer address 10.10.10.1 and VLAN 10 to it.

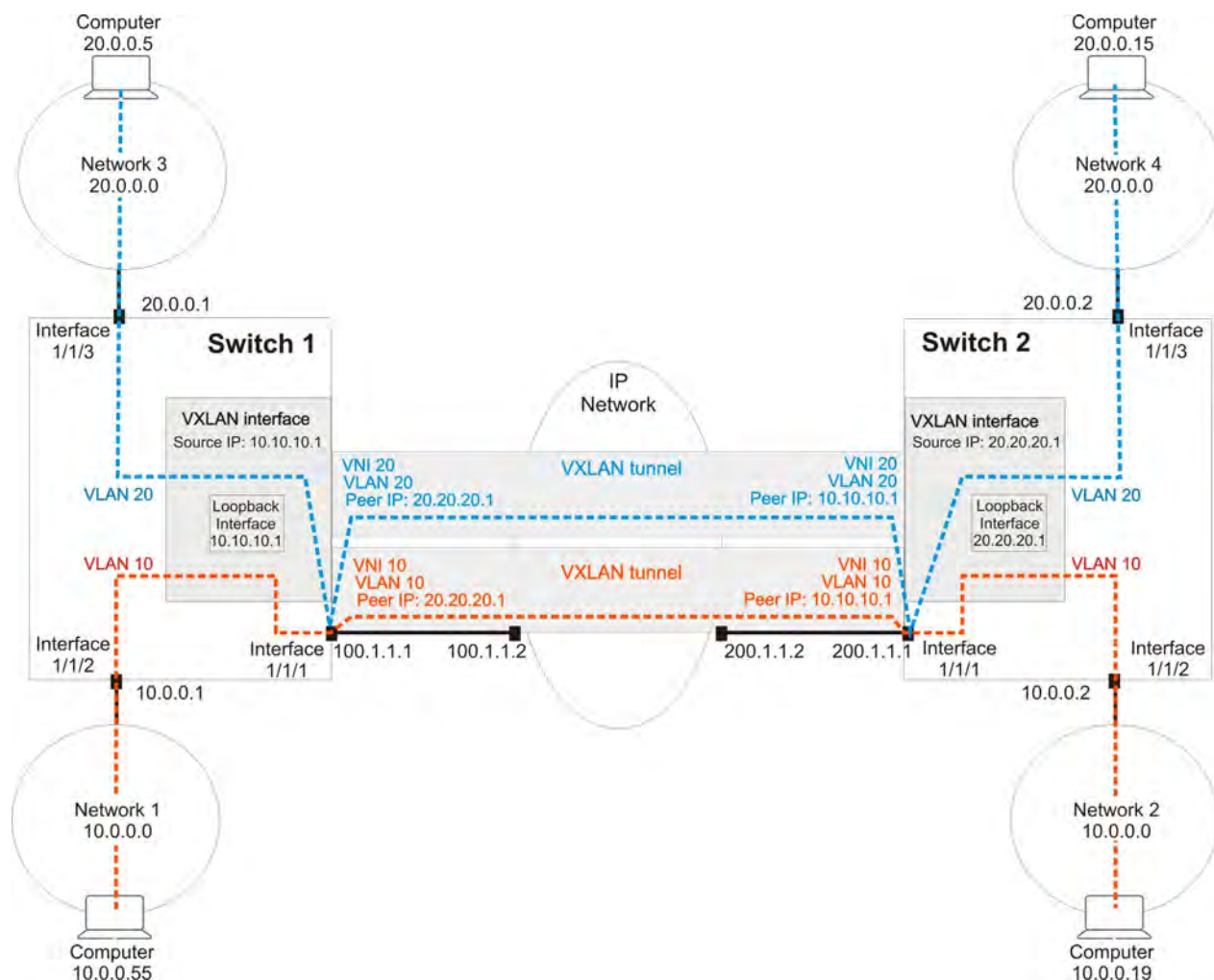
```
switch(config-vxlan-if)# vni 10
switch(config-vni)# vtep-peer 10.10.10.1
switch(config-vni)# vlan 10
switch(config-vni)# exit
switch(config-vxlan-if)# exit
switch(config)#
```

- g. Add a route to the peer VTEP peer address 10.10.10.1.

```
switch(config)# ip route 10.10.10.1 200.1.1.2
```

Static VXLAN with multiple VNIs between two VTEPs

This example creates two static VXLAN tunnels between two switches, enabling traffic from two networks to traverse underlay IP network.



Procedure

1. On switch 1:

- a. Enable loopback interface 1 and assign the IP address 10.10.10.1/24 to it.

```
switch# config
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 10.10.10.1/24
switch(config-if)# exit
switch(config)#
```

- b. Enable interface 1/1/1 and assign the IP address 100.1.1.1/24 to it.

```
switch# config
switch(config)# interface 1/1/1
switch(config-if)# routing (6300/6400 only)
switch(config-if)# ip address 100.1.1.1/24
switch(config-if)# no shutdown
switch(config-if)# exit
switch(config)#
```

- c. Create VLAN 10 and VLAN 20.

```
switch(config)# vlan 10,20
switch(config-vlan-10)# exit
```

- d. Enable interface 1/1/2 and assign VLAN 10 to it.

```
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# no routing (8325/8360/8400 only)
switch(config-if)# vlan access 10
switch(config-if)# exit
switch(config)#
```

- e. Enable interface 1/1/3 and assign VLAN 20 to it.

```
switch(config)# interface 1/1/3
switch(config-if)# no shutdown
switch(config-if)# no routing (8325/8360/8400 only)
switch(config-if)# vlan access 20
switch(config-if)# exit
switch(config)#
```

- f. Create VXLAN interface 1 and assign the source IP address 10.10.10.1 to it.

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# source ip 10.10.10.1
switch(config-vxlan-if)# no shutdown
```

- g. Create VNI 10, and assign the VTEP peer address 20.20.20.1 and VLAN 10 to it.

```
switch(config-vxlan-if)# vni 10
switch(config-vni)# vtep-peer 20.20.20.1
switch(config-vni)# vlan 10
```



```
switch(config-vni) # exit  
switch(config-vxlan-if) # exit
```

- h.** Create VNI 20, and assign the VTEP peer address 20.20.20.1 and VLAN 20 to it.

```
switch(config-vxlan-if) # vni 20  
switch(config-vni) # vtep-peer 20.20.20.1  
switch(config-vni) # vlan 20  
switch(config-vni) # exit  
switch(config-vxlan-if) # exit  
switch(config) #
```

2. On switch 2:

- a.** Enable loopback interface 2 and assign the IP address 20.20.20.1/24 to it.

```
switch# config  
switch(config) # interface loopback 2  
switch(config-loopback-if) # ip address 20.20.20.1/24  
switch(config-if) # exit  
switch(config) #
```

- b.** Enable interface 1/1/1 and assign the IP address 200.1.1.1/24 to it.

```
switch# config  
switch(config) # interface 1/1/1  
switch(config-if) # routing (6300/6400 only)  
switch(config-if) # ip address 200.1.1.1/24  
switch(config-if) # no shutdown  
switch(config-if) # exit  
switch(config) #
```

- c.** Create VLAN 10 and VLAN 20.

```
switch(config) # vlan 10,20  
switch(config-vlan-10) # exit
```

- d.** Enable interface 1/1/2 and assign VLAN 10 to it.

```
switch(config) # interface 1/1/2  
switch(config-if) # no shutdown  
switch(config-if) # no routing (8325/8360/8400 only)  
switch(config-if) # vlan access 10  
switch(config-if) # exit  
switch(config) #
```

- e.** Enable interface 1/1/3 and assign VLAN 20 to it.

```
switch(config) # interface 1/1/3  
switch(config-if) # no shutdown  
switch(config-if) # no routing (8325/8360/8400 only)  
switch(config-if) # vlan access 20  
switch(config-if) # exit  
switch(config) #
```

- f.** Create VXLAN interface 1 and assign the source IP address 20.20.20.1 to it.

```
switch(config) # interface vxlan 1  
switch(config-vxlan-if) # source ip 20.20.20.1  
switch(config-vxlan-if) # no shutdown
```

- g.** Create VNI 10, and assign the VTEP peer address 10.10.10.1 and VLAN 10 to it.

```
switch(config-vxlan-if) # vni 10
switch(config-vni) # vtep-peer 10.10.10.1
switch(config-vni) # vlan 10
switch(config-vni) # exit
switch(config-vxlan-if) # exit
```

- h.** Create VNI 20, and assign the VTEP peer address 10.10.10.1 and VLAN 20 to it.

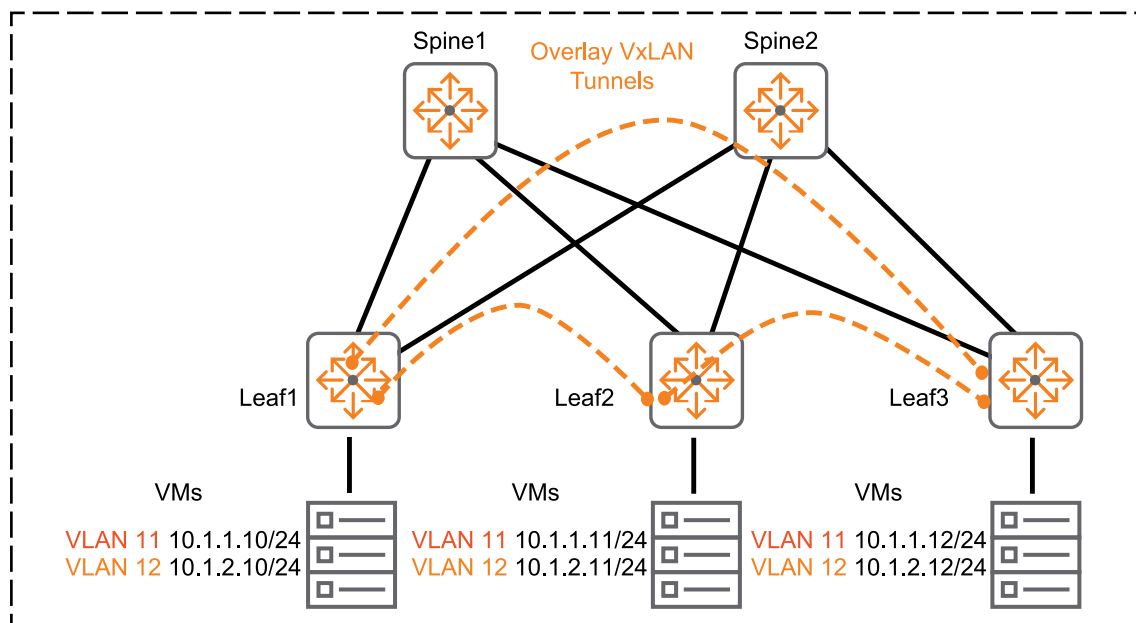
```
switch(config-vxlan-if) # vni 20
switch(config-vni) # vtep-peer 10.10.10.1
switch(config-vni) # vlan 20
switch(config-vni) # exit
switch(config-vxlan-if) # exit
switch(config) #
```

(Not supported on the 6200 Switch Series.)

Ethernet VPN (EVPN) is used as the overlay control plane to exchange layer 2 and layer 3 connectivity information between different layer 2/3 domains over an IP or IP/MPLS underlay network.

The initial EVPN standard RFC 7432 defined the BGP EVPN control plane and specifies an MPLS data-plane. The control plane with an MPLS data plane was extended to consider additional data plane encapsulations models. These models include VXLAN, NVGRE, and MPLS over GRE which is detailed in RFC 8365. This section focuses on EVPN and its operation with a VXLAN data plane for building overlay networks.

EVPN, as an overlay, supports multi-tenancy and is highly extensible, often using resources from different overlay fabric networks (like data centers) to deliver a single service. It can provide layer 2 connectivity over physical infrastructure for devices in a virtual network, or enable layer 3 routing.



In a Clos topology (leaf-spine) deployment of a Vxlan fabric, leaf nodes are configured as VTEPs to provide VXLAN services and spine nodes perform layer-3 routing on the outer headers IP/UDP encapsulation. If all VTEPs and underlay network devices of an EVPN network belong to the same AS, the spine nodes can act as route reflectors (RRs) to reflect routes between the VTEPs. In this scenario, the spine nodes receive and advertise BGP EVPN routes, but do not perform VXLAN encapsulation and de-encapsulation. If eBGP is used, the spines will forward the routes from one leaf to the other as inherent functionality of eBGP.

The IETF specifies two methods for supporting inter-subnet routing with EVPN: symmetric and asymmetric integrated routing and bridging (IRB). The main difference between the two methods is that the symmetric method supports both routing and bridged on both the ingress and egress VTEPs, where the asymmetric method supports routing on the ingress, but only bridging on the egress.

AOS-CX supports Centralized L3 gateways and Symmetric IRB Distributed L3 gateways.

Ethernet VPN (EVPN)-based VXLAN overview

Ethernet VPN (EVPN) is a standards-based BGP control plane to advertise MAC addresses, MAC/IP bindings, IP Prefixes, and so on. The initial EVPN standard RFC 7432 defined the BGP EVPN control plane and specifies an MPLS data-plane. The control plane with an MPLS data plane was extended to consider additional data plane encapsulations models. These models include VXLAN, NVGRE, and MPLS over GRE which is detailed in RFC 8365. This section focuses on EVPN and its operation with a VXLAN data plane for building overlay networks.

Static VXLAN uses flood and learn (or ingress replication) to learn the MACs of the remote host, which involves manual configurations of remote VXLAN Tunnel End Points (VTEPs) in the flood list. MP-BGP EVPN is used to discover remote VTEPs and advertise MAC address and MAC/IP bindings in the VXLAN overlay, thus eliminating the flood and learn. MP-BGP supports a new EVPN Network Layer Reachability Information (NLRI) carried in BGP using Multiprotocol BGP Extensions with a newly defined Address Family Identifier (AFI) and Subsequent Address Family Identifier (SAFI). Route Distinguisher (RD) is a unique number prepended to the advertised address within the VRF, ensuring support for overlapping IP addresses and MACs across different tenants. Routes can be imported and exported across VLANs and VRFs using a BGP extended community called Route Target (RT) that are advertised along with the EVPN routes.



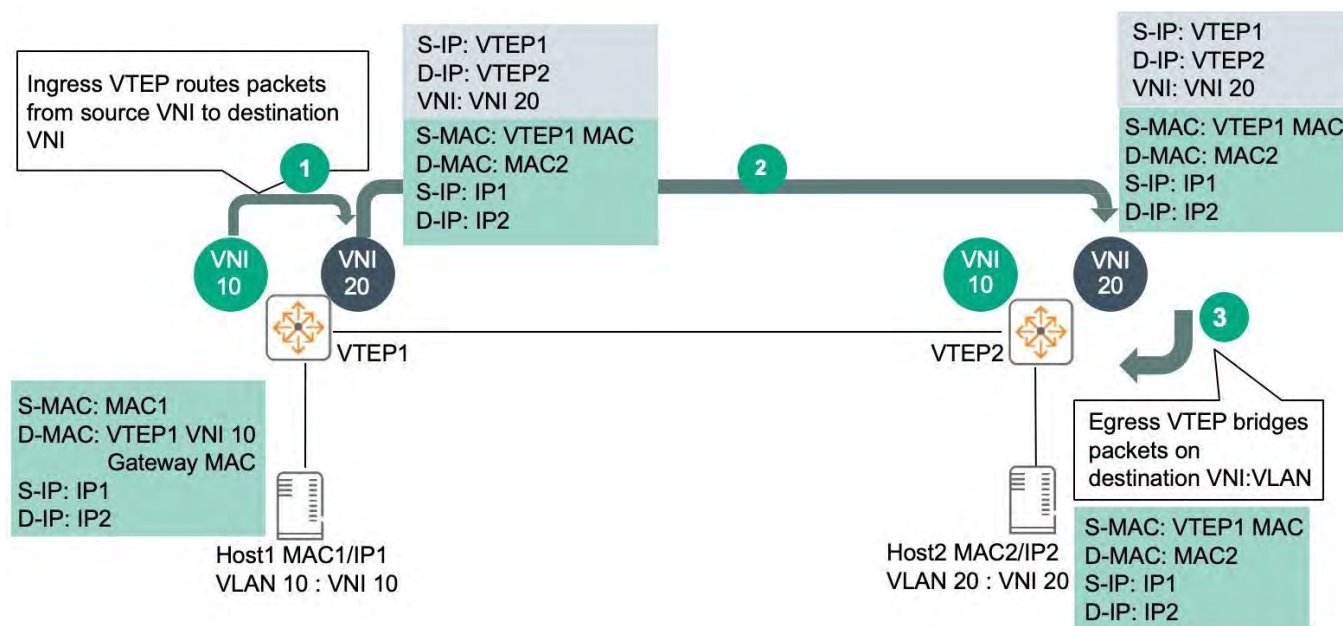
NOTE: Configuring static VTEPs is not supported when EVPN is enabled.

VMware NSX-V integration and EVPN are mutually exclusive and cannot be configured together. EVPN control pane is supported on the 6300, 6400, 8325, 8360, and 8400 switches.

See also *Border Gateway Protocol (BGP)* in the *IP Routing Guide*.

Asymmetric IRB

Asymmetric IRB uses different VNIs for bi-directional traffic between 2 hosts on different Layer-2 VNIs. For example:



1. Host1 in VLAN 10/VNI 10 connected to VTEP1 sends traffic to Host2 in VLAN 20/VNI 20 connected to VTEP2.
 - a. Traffic from Host1 is sent to VTEP1 VLAN 10/ VNI 10 gateway MAC.
2. VTEP1 routes traffic to VNI 20, encapsulates frame with VXLAN, adds outer Source/Destination IP, VNI info and sends traffic to VTEP2.
 - a. VTEP1 should already have a MAC/ARP entry for Host2.
 - b. The inner Source MAC is changed to VTEP1 VLAN 20 gateway MAC and inner Destination MAC is changed to MAC2 which belongs to Host2.
3. VTEP2 decapsulates outer VXLAN, bridges and sends the traffic to Host2 on VLAN 20/VNI 20.
 - a. Host2 will see source MAC as VTEP1.

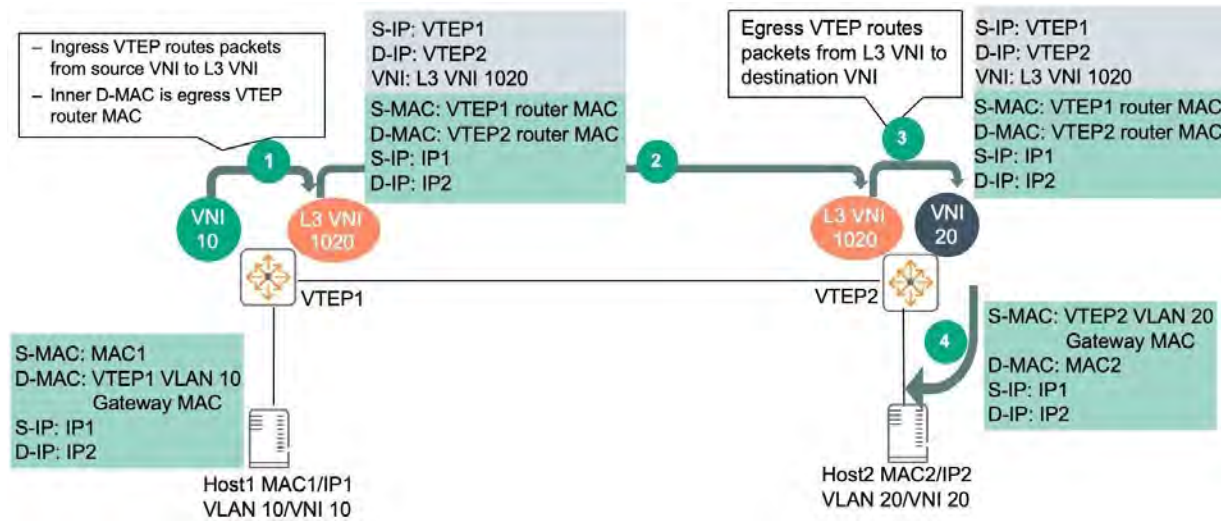
Return traffic from Host2 to Host1 is similar. Traffic from Host2 is sent to VTEP2 VLAN20/VNI 20 gateway MAC. VTEP2 routes traffic to VNI 10, encapsulates and sends the traffic to VTEP1. VTEP1 will decapsulate, bridge and send the traffic to Host1 on VLAN 10/VNI 10.

As seen from the traffic flow:

- Asymmetric IRB needs both source and destination Layer-2 VNIs to be configured on the ingress VTEP.
- Routing and bridging is used on the ingress VTEP.
- Bridging is used on the egress VTEP.
- Bi-directional traffic uses asymmetric paths:
 - Host1 to Host2 uses VNI 10 -> VNI 20.
 - Host2 to Host1 uses VNI 20 -> VNI 10.
- Asymmetric IRB will lead to increased ARP/MAC scale on VTEPs as they need to contain MAC/ARP of both source/destination hosts.
- If Asymmetric IRB is used, all subnets/VNIs have to be configured on all VTEPs. As previously shown, it is not mandatory for a subnet to span across all VTEPs in both Data Center and Campus networks, e.g. 10.10.220.0/24 only exists on Leaf1A/1B, Leaf2A/2B in Figure 2, 10.10.220.0/24 only exists in Building#1.

Symmetric IRB

Symmetric IRB uses the same Layer-3 VNI for bi-directional traffic between two hosts on different VLANs/ VNIs. For example:



1. Host1 in VLAN 10/VNI 10 connected to VTEP1 sends traffic to Host2 in VLAN 20/VNI 20 connected to VTEP2.
 - a. Traffic from Host1 is sent to VTEP1 VLAN 10/ VNI 10 gateway MAC.
2. VTEP1 routes traffic in the VRF (mapping to L3-VNI 1020), encapsulates traffic with VXLAN, adds outer Source/Destination IP, VNI info and sends traffic to VTEP2.
 - a. VTEP1 does not need to have a MAC/ARP entry for Host2.
 - b. VTEP1 learns the prefix route and/or host route corresponding to Host2 via EVPN Route type 5 or 2 respectively thus facilitating routing to the destination.
 - c. The inner Source MAC is changed to VTEP1 router MAC and inner Destination MAC is changed to VTEP2 router MAC.
3. VTEP2 decapsulates outer VXLAN, routes and sends the traffic to Host2 on VLAN 20/VNI 20.
 - a. Host2 will see source MAC as VTEP2 gateway MAC.

Return traffic from Host2 to Host1 is similar, traffic from Host2 is sent to VTEP2 VLAN20/VNI 20 gateway MAC. VTEP2 routes traffic to L3 VNI 1020, encapsulates and sends the traffic to VTEP1. VTEP1 decapsulates, route and send the traffic to Host1 on VLAN 10/VNI 10.

As seen from the traffic flow:

- Routing is used on both ingress and egress VTEPs.
- Bi-directional traffic uses symmetric path; same L3 VNI in both directions.
- VTEPs do not need to hold unnecessary ARP/MAC resources.
- Destination VLAN/VNI does not have to be configured on source VTEP.

Example of external connectivity and IVRL with symmetric IRB

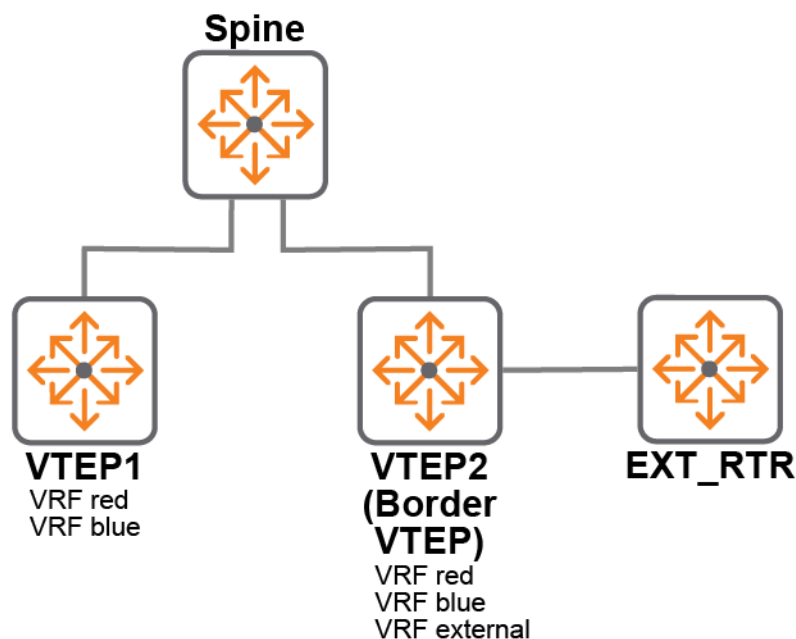
In symmetric IRB deployment, IVRL is typically configured at the border VTEPs.

At the border VTEP, host routes (published by remote VTEPs via route type-2) are installed. Similarly, the EVPN type-5 prefixes advertised by other VTEPs get installed as prefix-routes (for example, a /24 prefix for IPv4 overlay network). These routes can further be aggregated and leaked to a different VRF (for example, a VRF named **external**) and then be advertised to external network via protocols such as OSPF or EBGp.

Similarly, at the border VTEP, the routes at VRF **external** can be leaked to EVPN overlay VRFs and then be advertised as type-5 prefixes to all other VTEPs.

Example

This example shows the topology and sample configuration where IVRL is used along with symmetric IRB:



There are 2 VTEPs, namely VTEP1 and VTEP2. VTEP2 is the Border VTEP. Both the VTEPs have two tenant VRFs, namely VRF **red** and VRF **blue**. At VTEP2, there is an additional VRF **external** configured to have an external connectivity with EXT_RTR. EXT_RTR is a router running OSPF. At VTEP2, OSPF routes received from EXT_RTR at VRF **external** are leaked to VRF **red** and VRF **blue** (by configuring `redistribute ospf` in the `router bgp` context and configuring route targets required for route leaking), and then the leaked routes are advertised as EVPN type 5 prefixes to VTEP1.

Also at VTEP2, EVPN overlay routes in VRF **red** and VRF **blue** are leaked to VRF **external** and then the leaked routes are advertised by OSPF (by configuring `redistribute bgp` in the `router ospf` context) to the external router.

```
SPINE:
-----
!
router ospf 1
  router-id 2.2.2.2
  area 0.0.0.0
interface 1/1/1
  no shutdown
  routing
```

```

        ip address 10.10.10.2/24
        ip ospf 1 area 0.0.0.0
interface 1/1/2
    no shutdown
    routing
    ip address 20.20.20.2/24
    ip ospf 1 area 0.0.0.0
interface loopback 1
    ip address 2.2.2.2/24
    ip ospf 1 area 0.0.0.0
router bgp 1
    neighbor 1.1.1.1 remote-as 1
    neighbor 1.1.1.1 update-source loopback 1
    neighbor 3.3.3.3 remote-as 1
    neighbor 3.3.3.3 update-source loopback 1
    address-family l2vpn evpn
        neighbor 1.1.1.1 activate
        neighbor 1.1.1.1 route-reflector-client
        neighbor 1.1.1.1 send-community extended
        neighbor 3.3.3.3 activate
        neighbor 3.3.3.3 route-reflector-client
        neighbor 3.3.3.3 send-community extended
    exit-address-family

LEAF1:
-----
vrf blue
    rd 1:100
    route-target export 1:100 evpn
    route-target import 1:100 evpn
vrf red
    rd 1:200
    route-target export 1:200 evpn
    route-target import 1:200 evpn
!
vlan 1,20,40
virtual-mac 00:01:00:00:00:00
!
router ospf 1
    router-id 1.1.1.1
    area 0.0.0.0
evpn
    vlan 20
        rd auto
        route-target export auto
        route-target import auto
    vlan 40
        rd auto
        route-target export auto
        route-target import auto
interface 1/1/1
    no shutdown
    routing
    ip address 10.10.10.1/24
    ip ospf 1 area 0.0.0.0
interface 1/1/2
    no shutdown
    no routing
    vlan access 20
interface 1/1/3
    no shutdown
    no routing
    vlan access 40
interface loopback 1

```



```

    ip address 1.1.1.1/24
    ip ospf 1 area 0.0.0.0
interface vlan20
    vrf attach blue
    ip address 100.100.20.2/24
    active-gateway ip mac 00:01:00:00:00:01
    active-gateway ip 100.100.20.1
interface vlan40
    vrf attach red
    ip address 200.200.40.2/24
    active-gateway ip mac 00:01:00:00:00:01
    active-gateway ip 200.200.40.1
interface vxlan 1
    source ip 1.1.1.1
    no shutdown
    vni 20
        vlan 20
    !
    vni 40
        vlan 40
    !
    vni 10000
        routing
        vrf blue
    vni 20000
        routing
        vrf red
router bgp 1
    neighbor 2.2.2.2 remote-as 1
    neighbor 2.2.2.2 update-source loopback 1
    address-family l2vpn evpn
        neighbor 2.2.2.2 activate
        neighbor 2.2.2.2 send-community extended
    exit-address-family
!
    vrf blue
        address-family ipv4 unicast
            redistribute connected
        exit-address-family
!
    vrf red
        address-family ipv4 unicast
            redistribute connected
        exit-address-family
!

LEAF2: (Border leaf)
-----
vrf blue
    rd 1:100
    route-target export 1:100 evpn
    route-target import 1:100 evpn
    address-family ipv4 unicast
        route-target export 1:100
        route-target import 1:300
    exit-address-family
vrf red
    rd 1:200
    route-target export 1:200 evpn
    route-target import 1:200 evpn
    address-family ipv4 unicast
        route-target export 1:200
        route-target import 1:300
    exit-address-family

```

```

vrf external
  rd 1:300
  address-family ipv4 unicast
    route-target export 1:300
    route-target import 1:100
    route-target import 1:200
  exit-address-family
vlan 1,10,30
virtual-mac 00:02:00:00:00:00
!
router ospf 1
  router-id 3.3.3.3
  area 0.0.0.0
router ospf 1 vrf external
  router-id 30.1.1.1
  redistribute bgp
  area 0.0.0.0
evpn
  vlan 10
    rd auto
    route-target export auto
    route-target import auto
  vlan 30
    rd auto
    route-target export auto
    route-target import auto
interface 1/1/1
  no shutdown
  routing
  ip address 20.20.20.1/24
  ip ospf 1 area 0.0.0.0
interface 1/1/2
  no shutdown
  no routing
  vlan access 10
interface 1/1/3
  no shutdown
  no routing
  vlan access 30
interface 1/1/4
  no shutdown
  routing
  vrf attach shared
  ip address 30.1.1.1/24
  ip ospf 1 area 0.0.0.0
interface loopback 1
  ip address 3.3.3.3/24
  ip ospf 1 area 0.0.0.0
interface vlan10
  vrf attach blue
  ip address 100.100.10.2/24
  active-gateway ip mac 00:02:00:00:00:02
  active-gateway ip 100.100.10.1
interface vlan30
  vrf attach red
  ip address 200.200.30.2/24
  active-gateway ip mac 00:02:00:00:00:02
  active-gateway ip 200.200.30.1
interface vxlan 1
  source ip 3.3.3.3
  no shutdown
  vni 10
  vlan 10
!
```

```

vni 30
    vlan 30
!
vni 10000
    routing
    vrf blue
!
vni 20000
    routing
    vrf red
!
router bgp 1
    neighbor 2.2.2.2 remote-as 1
    neighbor 2.2.2.2 update-source loopback 1
    address-family l2vpn evpn
        neighbor 2.2.2.2 activate
        neighbor 2.2.2.2 send-community extended
    exit-address-family
!
    vrf blue
        address-family ipv4 unicast
            redistribute connected
        exit-address-family
!
    vrf red
        address-family ipv4 unicast
            redistribute connected
        exit-address-family
!
    vrf external
        address-family ipv4 unicast
            redistribute connected
            redistribute ospf
        exit-address-family
!

EXT_RTR(OSPF): Router connected to border leaf
-----
vrf external
router ospf 1 vrf external
    router-id 30.1.1.2
    area 0.0.0.0
interface 1/1/1
    no shutdown
    routing
    vrf attach external
    ip address 30.1.1.2/24
    ip ospf 1 area 0.0.0.0
interface 1/1/2
    no shutdown
    routing
    vrf attach external
    ip address 40.1.1.1/24
    ip ospf 1 area 0.0.0.0

```



NOTE: The locally learned neighbor entries (ARP/ND) on a VTEP, dynamically learned (via data plane) over client/tenant-facing SVI, can be published as host routes via Route Type-2 only over the BGP-EVPN control plane. These dynamic entries cannot be redistributed as host routes in a non BGP-EVPN routing domain. For example, in the figure above, `redistribute host-route` is configured under the EVPN-VLAN context to ensure that locally learned ARP entries at VTEP2 are advertised in Route Type-2 as /32 prefixes to VTEP1. However, these ARP entries cannot be advertised as /32 routes to EXT_RTR.

MP-BGP extension for EVPN

The following are some important EVPN route types:

- **Ethernet auto-discovery route (Route 1)**—Advertises Ethernet Segment (ES) information in multihomed sites.
- **MAC/IP advertisement route (Route 2)**—Advertises MAC, MAC/IP(v6) bindings (mapping to neighbor entry and/or host route) of the locally learned host.
- **Inclusive Multicast Ethernet tag (IMET) route (Route 3)**—Advertises VTEP and VXLAN mappings for automating VTEP discovery, VXLAN tunnel establishment, and VXLAN tunnel assignment.
- **Ethernet Segment route (Route 4)**—Advertises ES and VTEP mappings.
- **IP Prefix advertisement route (Route 5)**—Advertises BGP IPv4 or IPv6 unicast routes as IP prefixes.

MP-BGP uses the route distinguisher (RD) field to differentiate BGP EVPN routes of different L2 domains and uses route targets to control the advertisement and acceptance of BGP EVPN routes. MP-BGP supports the following types of route targets:

- **Export Route Target**—A VTEP sets the export targets for BGP EVPN routes learned from the local site before advertising them to remote VTEPs.
- **Import Route Target**—A VTEP which receives the EVPN routes imports them if any of the route targets in the received route matches with any of the locally configured import route targets.



NOTE: The current release supports only MAC and MAC/IP advertisement (route type 2), Inclusive multicast Ethernet tag (IMET) routes (route type 3), and IP prefix route advertisement (route type 5).

Auto-discovery via EVPN

VTEPs use BGP EVPN routes to discover VTEP neighbors, establish VXLAN tunnels, and assign the tunnels to VNIs.

- **IMET route**—VTEPs advertise their locally configured L2 VNIs through IMET routes. Two VTEPs configured with the same L2 VNI provision VXLAN tunnel mapping to remote VTEP and L2 VNI. IMET is leveraged to create the replication list for the forwarding of Broadcast, Unknown Unicast, and Multicast (BUM) traffic.
- **MAC/IP advertisement route and IP prefix advertisement route**—In the EVPN deployment, VTEPs advertise MAC/IP advertisement routes or IP prefix advertisement routes which are imported by the other VTEPs based on route targets.

Layer 2 forwarding

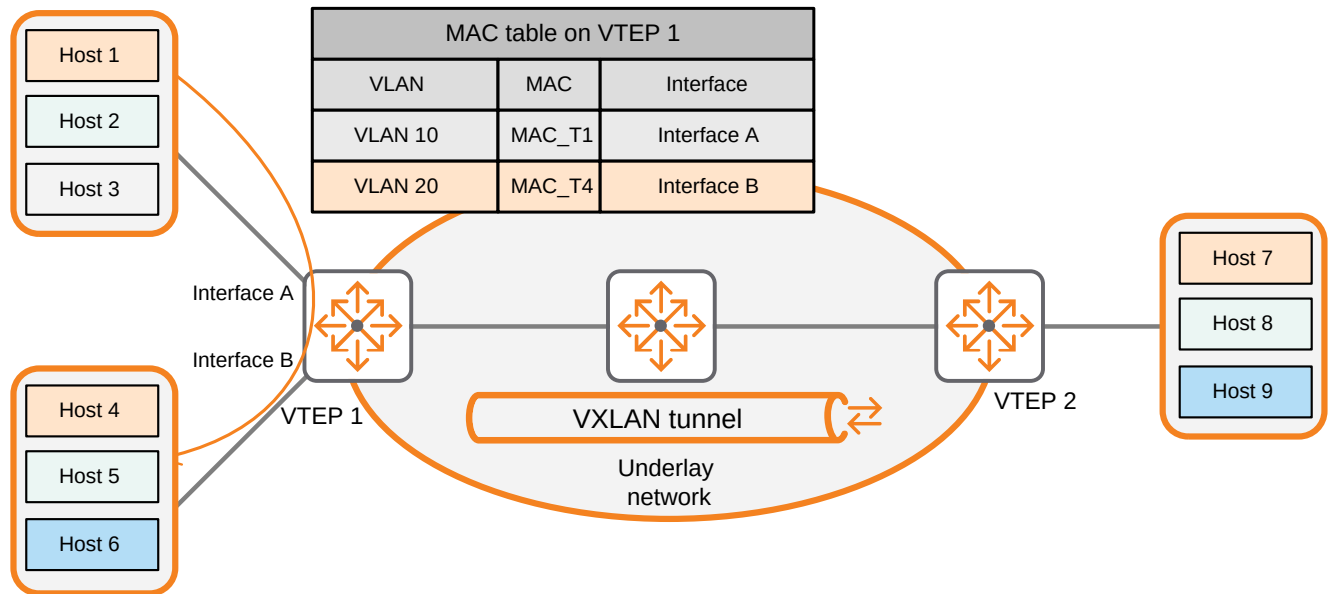
MAC learning

The VTEP learns MAC addresses by using the following methods:

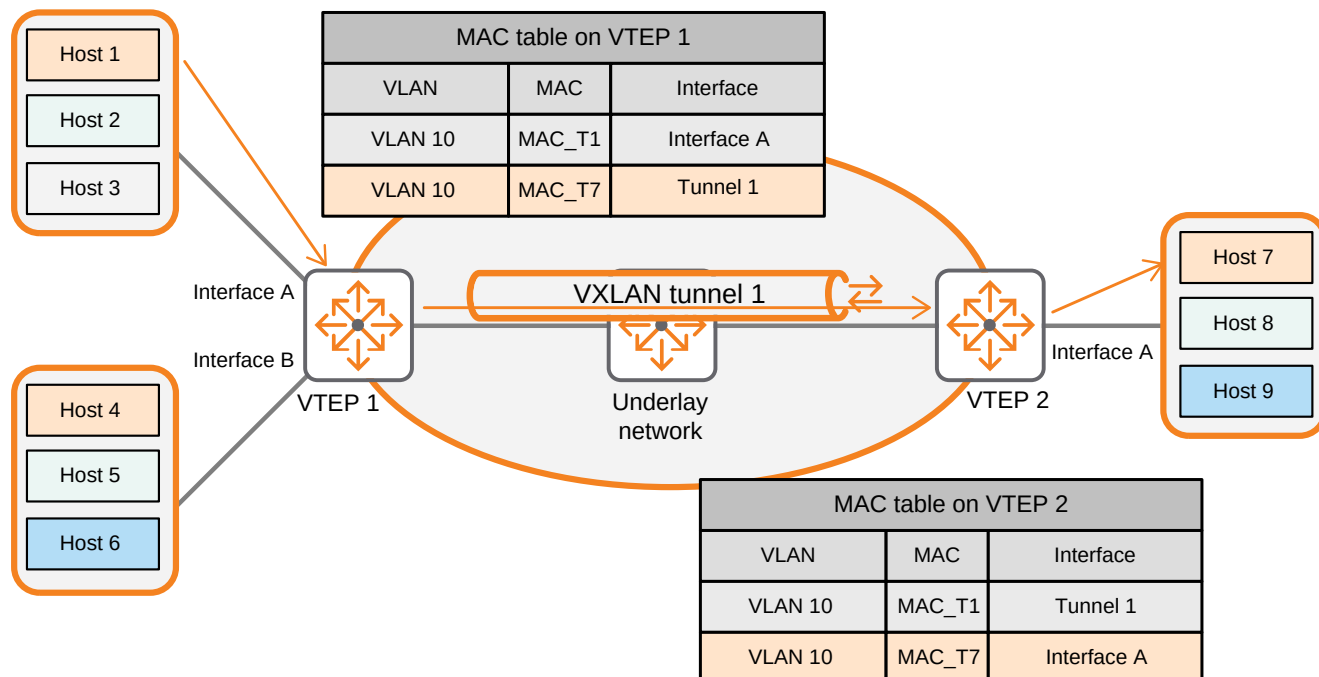
- **Local MAC learning**—The VTEP automatically learns the source MAC addresses of frames sent from the local site. The outgoing interfaces of local MAC address entries are interfaces on which the MAC addresses are learned.
- **Remote MAC learning**—The VTEP uses MP-BGP to advertise local MAC reachability information to remote VTEPs and learn MAC reachability information from remote VTEPs. The outgoing interfaces of MAC address entries advertised from a remote site are VXLAN tunnel interfaces. A remote host's MAC address entries are not learned from the dataplane over the VXLAN fabric.

Unicast

In the following example, VTEP 1 performs Layer 2 forwarding for local MAC learning and forwarding:



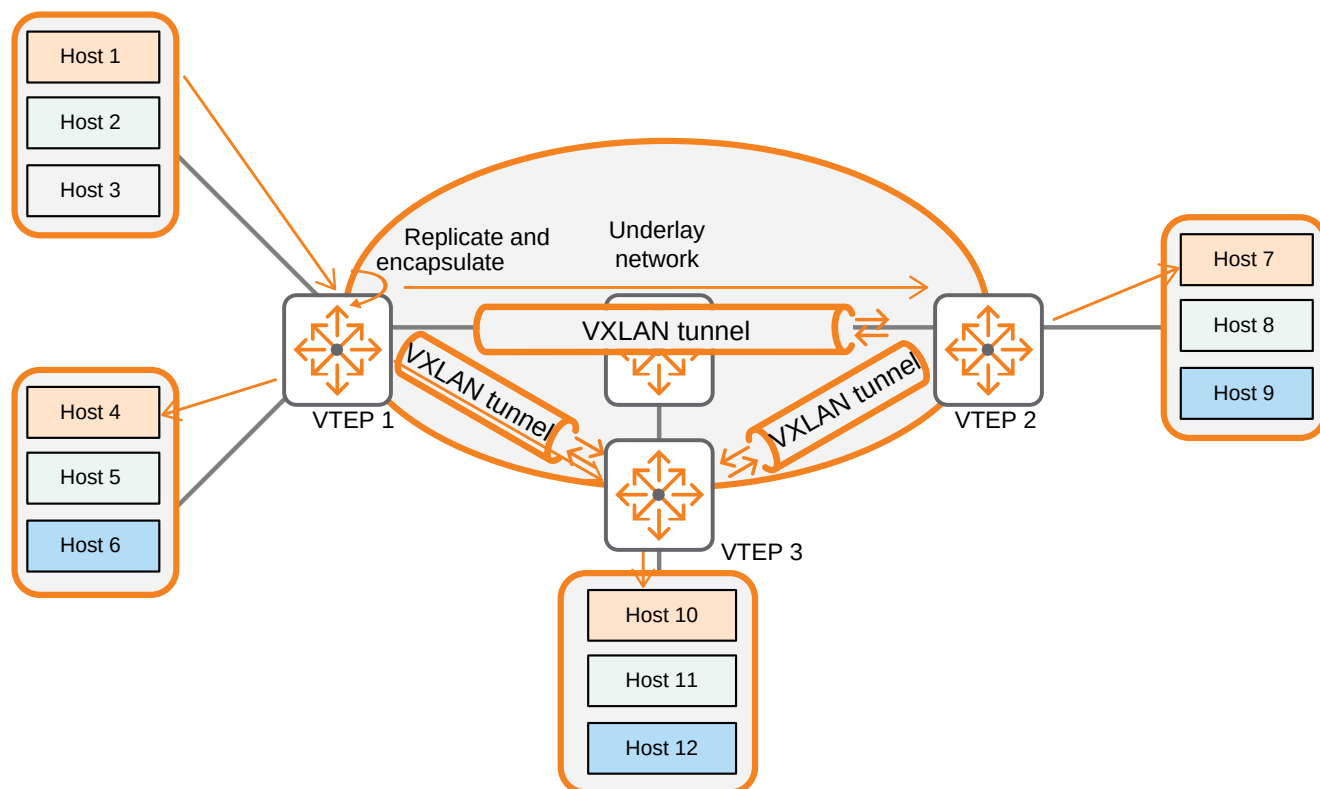
In the following example, VTEP 1 performs remote MAC learning and forwarding:



The following process applies to remote MAC learning and forwarding:

1. The source VTEP encapsulates the Ethernet frame in the VXLAN/UDP/IP header.
In the outer IP header, the source IP address is the source VTEP's VXLAN tunnel source IP address. The destination IP address is the VXLAN tunnel destination IP address.
2. The source VTEP forwards the encapsulated packet out of the outgoing VXLAN tunnel interface.
3. The intermediate underlay devices forward the packet to the destination VTEP by using the outer IP header.
4. The destination VTEP removes the headers on top of the inner Ethernet frame. It then performs MAC address table lookup in the VXLAN's VNI to forward the frame out of the matching outgoing interface.

Flood



A VTEP floods a Broadcast, Unknown Unicast, and Multicast (BUM) traffic to all local interfaces excluding the incoming interface. The source VTEP also replicates the flood frame, and then sends one replica to the destination IP address of each VXLAN tunnel in the VXLAN. Each destination VTEP floods the inner Ethernet frame to all the host/tenant facing interfaces. To avoid loops, the default VXLAN split horizon rule applies where the destination VTEPs do not flood the frame to VXLAN tunnels.



NOTE: For BUM traffic flooding, only Head End Replication (HER) is supported.

Layer 3 forwarding

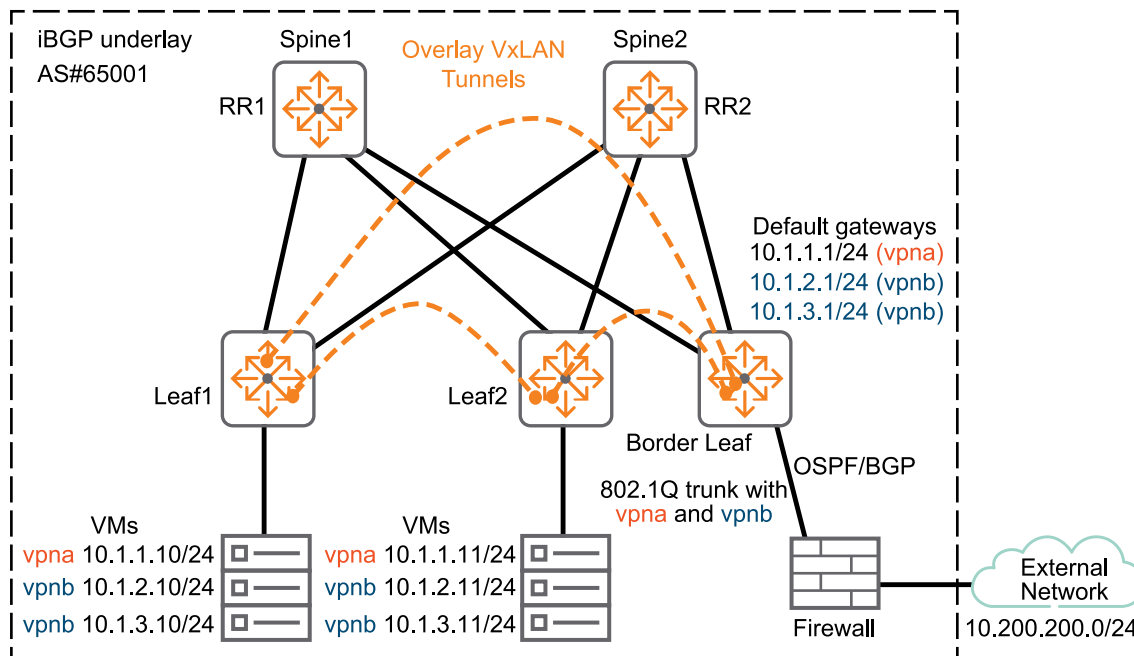
EVPN uses either Centralized L3 gateway or Distributed L3 gateways to provide Layer 3 forwarding services for hosts in VXLANs.

- **Centralized L3 gateway:** Uses one VTEP to provide Layer 3 forwarding for VXLANs. Typically, the gateway-collocated VTEP connects to other VTEPs and the external network.
- **Distributed L3 gateway:** Deploys one EVPN gateway on each VTEP to provide optimized Layer 3 forwarding for directly connected hosts.

In either design, the gateways use virtual Layer 3 interfaces as gateway interfaces for VXLANs. An AOS-CX switch cannot participate in both Centralized L3 gateway and Distributed L3 gateway topologies at the same time.

Centralized L3 gateway deployment

In the following topology, a VTEP acts as a routing (Layer 3) gateway for hosts. This is the only VTEP in the VXLAN fabric, that performs Layer 3 forwarding (routing). All other VTEPs are configured to perform Layer 2 bridging.



The network uses the following process to forward Layer 3 traffic from a host to the destination:

1. The centralized gateway (VTEP) publishes the gateway MAC(s) for all Layer-3 Gateway SVIs (with a L2 VNI mapping), to all other VTEPs.
2. Traffic is triggered by VM in VPNA (for example, 10.1.1.10/24) to VM (for example, 10.1.2.11/24) behind remote VTEP in VPNB (see the topology diagram for network details).
3. The host sends an ARP request to obtain the MAC address of the SVI interface that acts as the gateway, and then sends the Layer 3 traffic to the Centralized L3 gateway.
4. The first hop VTEP looks up the matching MAC address table and forwards the traffic to the Centralized L3 gateway through a VXLAN tunnel.
5. The Centralized L3 gateway removes the VXLAN encapsulation and performs route lookup. In case the corresponding neighbor entry (ARP/ND) is found missing for the destination IP address, it performs neighbor resolution (ARP/ND) in the context Layer-2 VNI, that maps to the corresponding gateway SVI interface.
6. After resolving the neighbor entry, for subsequent packets received with same credentials, the centralized gateway performs VXLAN encapsulation of the packet with the destination VNI and forwards the packet to the destination VTEP.
7. Any return traffic follows the same set of procedures described above.



NOTE: From release 10.04 onwards, Active Gateway must be configured for Centralized L3 gateway deployments (both VSX and standalone).

Distributed L3 Anycast gateway

As the name suggests, all VTEPs, that are connected to hosts or clients belonging to same subnet, are configured with same gateway IP and MAC address (for the subnet), thus making this configuration anycast in nature. The (anycast) gateway IP and MAC address is configured on the client/tenant facing SVI interface.

Anycast gateway configuration is done using the `active-gateway` command under an SVI interface. For example:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# active-gateway ip 10.1.1.1 mac 00:00:02:02:02:02
```

Active-gateway configuration recommendations for an EVPN environment

- The active gateway virtual IP configuration must not be the same as the physical IP configuration of the interface.
- Same active gateway virtual IP and physical IP cannot be configured for an EVPN environment. If configured, it can lead to flooding of data traffic.
- Physical IP addresses configured on the SVI interface on VSX-primary and VSX-secondary must be different. (Note that the same rule would also apply if active gateway is configured on each VTEP to implement asymmetric IRB with distributed gateway.)
- The Active Gateway MAC for both IPv6 and IPv4 on the tenant SVI should be identical for Anycast Gateway configuration.
- The Active Gateway IPv6 address should be a link local address and identical on both VSX peers.

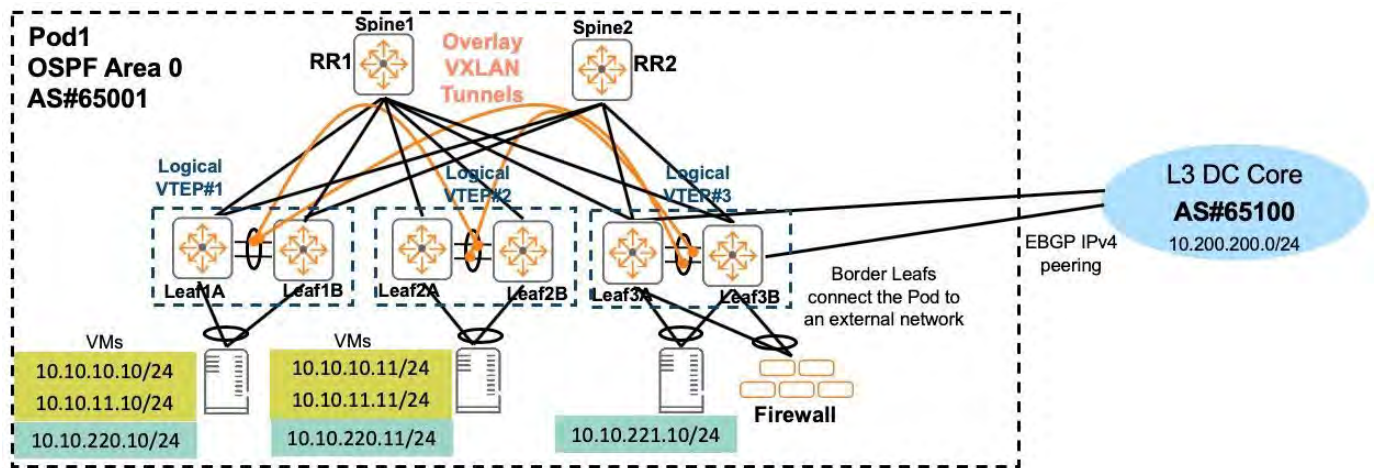
Symmetric IRB

Symmetric IRB involves routing and bridging on both ingress and egress VTEPs. Thus ensuring that bidirectional traffic is tagged with same L3VNI, indicating routing operation to be performed on the egress VTEP. Ingress VTEP routes the traffic between the source subnet and destination subnet, in a VRF (Virtual Routing and Forwarding), and sends the packet to the egress VTEP, encapsulated with L3VNI. Both ingress and egress VTEPs have the VRF mapped to the same L3VNI. The egress VTEP, routes the packet to, the destination subnet in the VRF (mapped to Layer3-VNI).

Symmetric IRB is an optimal solution in a large-scale network, as VTEPs can be selectively configured as gateway for the subnets present on the directly connected hosts.

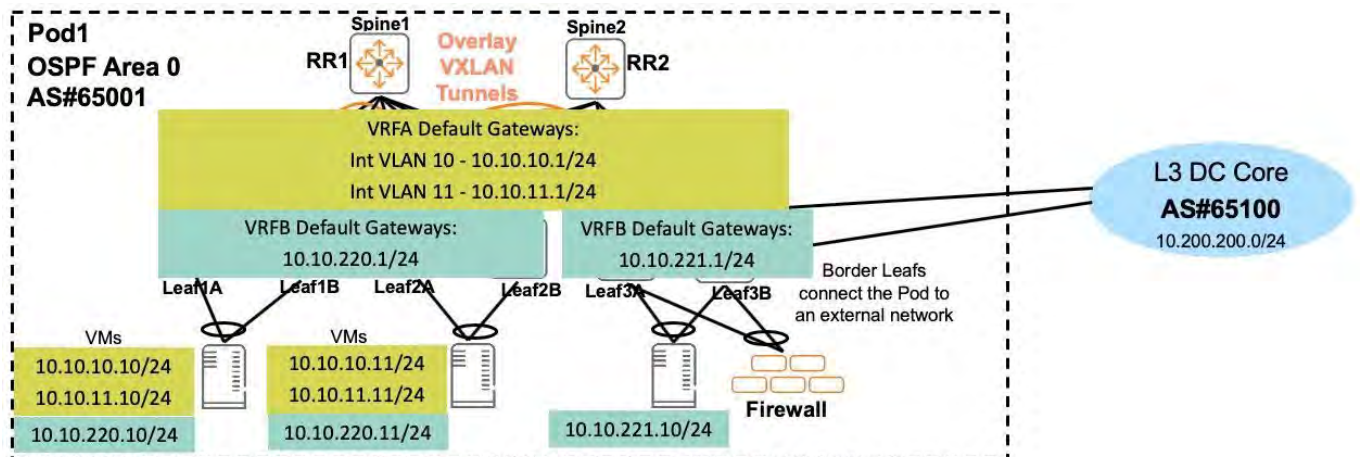
Symmetric IRB with distributed Anycast Gateway

Virtual Extensible LAN (VXLAN) tunnels created by Multi-Protocol BGP (MP-BGP) Ethernet VPN (EVPN) are typically deployed on VXLAN Tunnel End Points (VTEPs) to provide L2 network connectivity over a L3 Data Center network deployed in a Pod or Availability Zone. For example:



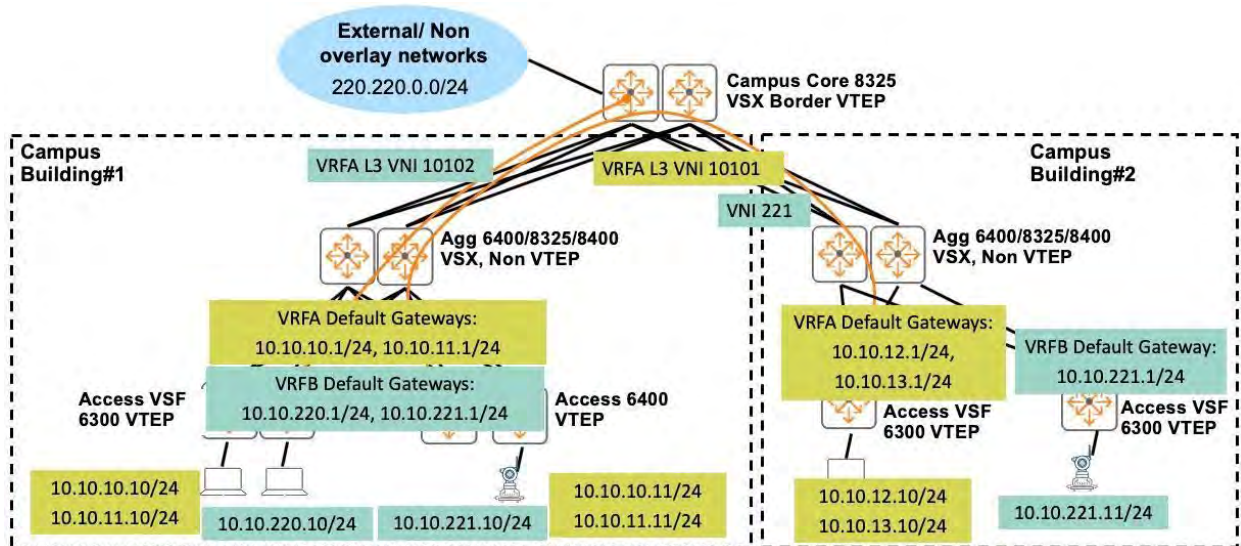
VSX Leaf switches in a rack form a logical VTEP and border leafs connect the EVPN domain to a L3 DC core via EBGP IPv4 peering. OSPF/IBGP EVPN is used within the Pod. Spine switches are configured as IBGP EVPN Route Reflectors (RRs), while Leaf VTEPs are configured as EVPN RR clients.

Distributed L3 Gateways can be deployed across the overlay network to optimize L3 network connectivity and minimize latency between Virtual Machines (VMs) within a rack as follows:



With Distributed L3 Gateways, the anycast 10.10.10.1/24 and 10.10.11.1/24 IPs are configured on Leaf1A/Leaf1B and Leaf2A/Leaf2B. Traffic between VMs 10.10.10.10/24 and 10.10.11.10/24 is routed directly on Leaf1A/Leaf1B, while traffic between VMs 10.10.10.11/24 and 10.10.11.11/24 is routed directly on Leaf2A/Leaf2B.

Distributed L3 Gateways can also be deployed in a campus network to optimize L3 network connectivity and minimize latency between clients connected to the same access switch. For example:



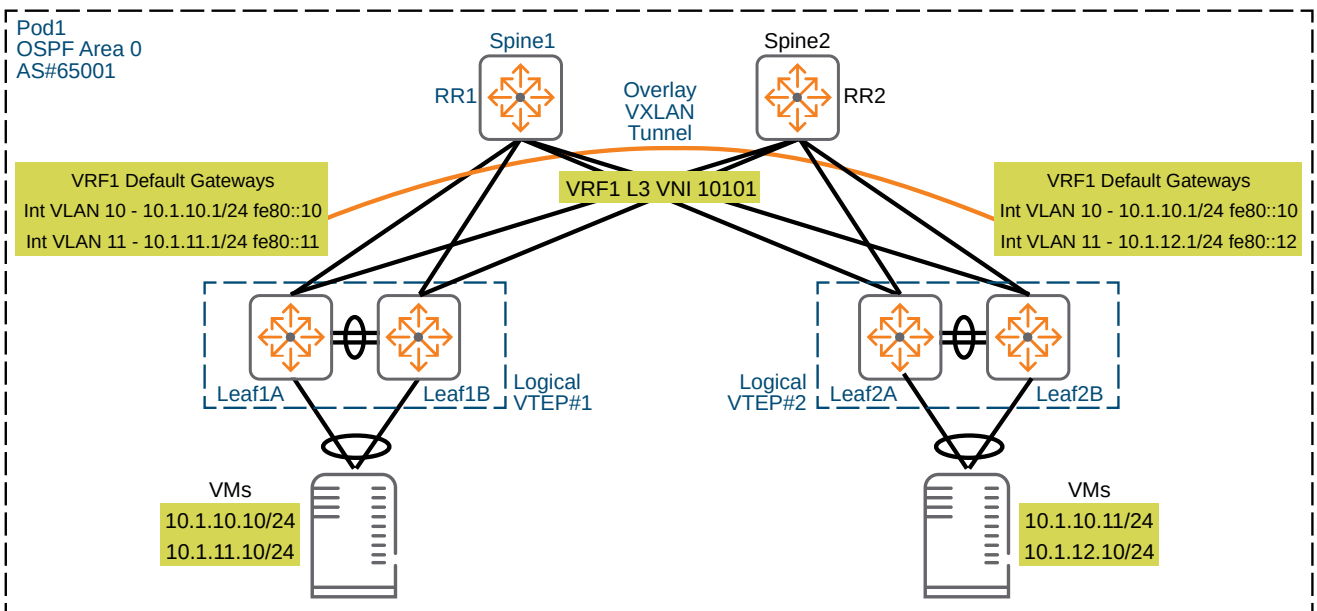
With Distributed L3 Gateways, the anycast 10.10.10.1/24 and 10.10.11.1/24 IPs are configured on access VSF 6300 and access 6400 in Campus building #1. Traffic between clients 10.10.10.10/24 and 10.10.11.10/24 is routed directly on access VSF 6300, while traffic between clients 10.10.10.11/24 and 10.10.11.11/24 is routed directly on access 6400.

VRFs can be added to VTEPs for L3 multi-tenancy across a VXLAN/EVPN overlay network.

In a campus, it is typical for different buildings to utilize different subnets. L3 VXLAN routing between different subnets on building #1 and building #2 would utilize the L3 VXLAN Network Identifier (VNI) assigned to its VRF. It is also possible to extend L2 VXLAN across buildings for IoT devices that need to remain on the same subnet.

VXLAN/EVPN symmetric IRB distributed L3 gateways example

This example provides an overview of a VXLAN/EVPN symmetric IRB distributed L3 gateways deployment.



Topology details

- Leaf1A/Leaf1B (logical VTEP#1) are in rack1, Leaf2A/Leaf2B (logical VTEP#2) are in rack2.
- VSX 8325, 8360 logical VTEPs within a rack utilize anycast Lo 1 IP as VXLAN tunnel source.
- VLAN 10 has distributed L3 gateways with anycast IP 10.1.10.1/24 and anycast IPv6 address fe80::10 in both racks.
- L2 VXLAN is used between VMs on different racks within VLAN 10/VNI 10.
- Optimized L3 network connectivity is provided for VMs in rack 1 between VLANs 10 and 11, this traffic does not need to leave the rack.
- Optimized L3 network connectivity is provided for VMs in rack 2 between VLANs 10 and 12, this traffic does not need to leave the rack.
- L3 VNI 10101 is only used for L3 network connectivity between different subnets/racks, e.g.
 - VLAN 10/VNI 10 in rack1 <-> VLAN12/VNI 12 in rack2
 - VLAN 11/VNI 11 in rack1 <-> VLAN12/VNI 12 in rack2
 - VLAN 11/VNI 11 in rack1 <-> VLAN10/VNI 10 in rack2

Sample spine 1 configuration

```

hostname Spine1
user admin group administrators password ciphertext AQBap!snip
!
!
ssh server vrf mgmt
!
interface mgmt
    no shutdown
    ip static 10.136.40.245/24
    default-gateway 10.136.40.6

! downlinks towards VTEPs
interface 1/1/25
    no shutdown
    mtu 9198
    routing
    ip mtu 9198
    ip address 192.168.3.0/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point
interface 1/1/26
    no shutdown
    mtu 9198
    routing
    ip mtu 9198
    ip address 192.168.3.2/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point
interface 1/1/27
    no shutdown
    mtu 9198
    routing
    ip mtu 9198
    ip address 192.168.3.4/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point
interface 1/1/28
    no shutdown
    mtu 9198

```

```

    routing
    ip mtu 9198
    ip address 192.168.3.6/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point

! Lo0 used for EVPN peering with VTEPs
interface loopback 0
    ip address 192.168.1.1/32
    ip ospf 1 area 0.0.0.0
!
!
router ospf 1
    router-id 192.168.1.1
    area 0.0.0.0
router bgp 65001
    bgp router-id 192.168.1.1
    neighbor leaf peer-group
    neighbor leaf remote-as 65001
    neighbor leaf description Leaf RR clients
    neighbor leaf update-source loopback 0
    neighbor 192.168.1.3 peer-group leaf
    neighbor 192.168.1.4 peer-group leaf
    neighbor 192.168.1.5 peer-group leaf
    neighbor 192.168.1.6 peer-group leaf
    address-family l2vpn evpn
        neighbor leaf route-reflector-client
        neighbor leaf send-community extended
        neighbor 192.168.1.3 activate
        neighbor 192.168.1.4 activate
        neighbor 192.168.1.5 activate
        neighbor 192.168.1.6 activate
! Lo0 used for EVPN peering with VTEPs
    exit-address-family
!
https-server vrf default
https-server vrf mgmt

```

Sample spine 2 configuration

```

hostname Spine2
user admin group administrators password ciphertext AQBap!snip
!
!
ssh server vrf mgmt
!
interface mgmt
    no shutdown
    ip static 10.136.40.246/24
    default-gateway 10.136.40.6

! downlinks towards VTEPs
interface 1/1/25
    no shutdown
    mtu 9198
    routing
    ip mtu 9198
    ip address 192.168.3.8/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point
interface 1/1/26
    no shutdown
    mtu 9198
    routing

```

```

    ip mtu 9198
    ip address 192.168.3.10/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point
interface 1/1/27
    no shutdown
    mtu 9198
    routing
    ip mtu 9198
    ip address 192.168.3.12/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point
interface 1/1/28
    no shutdown
    mtu 9198
    routing
    ip mtu 9198
    ip address 192.168.3.14/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point

! Lo0 used for EVPN peering with VTEPs
interface loopback 0
    ip address 192.168.1.2/32
    ip ospf 1 area 0.0.0.0
!
router ospf 1
    router-id 192.168.1.2
    area 0.0.0.0
router bgp 65001
    bgp router-id 192.168.1.2
    neighbor leaf peer-group
    neighbor leaf remote-as 65001
    neighbor leaf description Leaf RR clients
    neighbor leaf update-source loopback 0
    neighbor 192.168.1.3 peer-group leaf
    neighbor 192.168.1.4 peer-group leaf
    neighbor 192.168.1.5 peer-group leaf
    neighbor 192.168.1.6 peer-group leaf
    address-family l2vpn evpn
        neighbor leaf route-reflector-client
        neighbor leaf send-community extended
        neighbor 192.168.1.3 activate
        neighbor 192.168.1.4 activate
        neighbor 192.168.1.5 activate
        neighbor 192.168.1.6 activate
! Lo0 used for EVPN peering with VTEPs
    exit-address-family
!
https-server vrf default
https-server vrf mgmt

```

Sample Leaf1A Configuration

```

hostname Leaf1A
user admin group administrators password ciphertext AQBap!snip
vrf KA
vrf VRF1
    rd 65001:1
    route-target export 65001:1 evpn
    route-target import 65001:1 evpn
!
ssh server vrf mgmt
vlan 1
vlan 2
    vxsync
vlan 10
    vxsync

```



```

vlan 11
    vsx-sync
virtual-mac 02:00:00:00:01:00
! A unique virtual-mac is required on all symmetric IRB distributed L3 gateway VTEPs.
! It should be the same for a VSX logical VTEP pair, different virtual-macs should be used
! for other VSX VTEP pairs
! This is the inner D-MAC/egress VTEP router MAC

evpn
    vlan 10
        rd auto
        route-target export auto
        route-target import auto
        redistribute host-route
    vlan 11
        rd auto
        route-target export auto
        route-target import auto
        redistribute host-route
! redistribute host-route is used to advertise host routes (/32 for IPv4, /128 to IPv6) to remote VTEPs
! e.g. 10.1.10.100/32 and/or IPv6 address 1010::100/128 is connected to Leaf1A/Leaf1B
! 10.1.12.10/32 or IPv6 address 1012::10/128 is connected to Leaf2A/Leaf2B
! This is required as the same subnet is spread across multiple VTEPs
! Without host routes, a remote VTEP would not be aware of the correct destination VTEP
! to send traffic
!
interface mgmt
    no shutdown
    ip static 10.136.40.237/24
    default-gateway 10.136.40.6

! Southbound server facing ports
interface lag 1 multi-chassis
    no shutdown
    description Server
    no routing
    vlan trunk native 1
    vlan trunk allowed 10-11
    lacp mode active
interface 1/1/1
    no shutdown
    description Server
    lag 1

! Northbound uplink ports
interface 1/1/23
    no shutdown
    description Spine1
    mtu 9198
    ip mtu 9198
    ip address 192.168.3.1/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point
interface 1/1/24
    no shutdown
    description Spine2
    mtu 9198
    ip mtu 9198
    ip address 192.168.3.9/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point

! VSX ports
interface 1/1/41
    no shutdown
    vrf attach KA
    description VSX KeepAlive
    ip address 192.168.0.0/31
interface lag 256
    vsx-sync vlans
    no shutdown
    description ISL
    no routing
    vlan trunk native 1 tag
    vlan trunk allowed all
    lacp mode active
interface 1/1/49
    no shutdown
    mtu 9198
    description VSX ISL link
    lag 256
interface 1/1/50
    no shutdown
    mtu 9198
    description VSX ISL link

```

```

lag 256

! Unique Lo0 on each VTEP
interface loopback 0
    ip address 192.168.1.3/32
    ip ospf 1 area 0.0.0.0

! Shared anycast Lo1 between a VSX logical VTEP pair
interface loopback 1
    ip address 192.168.11.3/32
    ip ospf 1 area 0.0.0.0

! Transit VLAN between VSX nodes for IGP continuity
interface vlan 2
    ip mtu 9198
    ip address 192.168.3.200/31
    ip ospf 1 area 0.0.0.0
    ip ospf cost 50
    ip ospf network point-to-point

! In 10.6, SVI IP (and/or IPv6) address must be different from the anycast Active Gateway IP(and/or IPv6)address
! this provides easier troubleshooting while sourcing traffic from the SVI IP (and/or IPv6) that is not
! common to all VTEPs
! Configuration simplification with same IP address for SVI and Active Gateway is an
! enhancement for a future release.
interface vlan 10
    vsx-sync active-gateways
    vrf attach VRF1
    ip address 10.1.10.2/24
    ipv6 address 1010::2/64
    active-gateway ip mac 12:00:00:00:01:00
    active-gateway ip 10.1.10.1
    active-gateway ipv6 mac 12:00:00:00:01:00
    active-gateway ipv6 fe80::10
interface vlan 11
    vsx-sync active-gateways
    vrf attach VRF1
    ip address 10.1.11.2/24
    ipv6 address 1011::2/64
    active-gateway ip mac 12:00:00:00:01:00
    active-gateway ip 10.1.11.1
    active-gateway ipv6 mac 12:00:00:00:01:00
    active-gateway ipv6 fe80::11

interface vxlan 1
    source ip 192.168.11.3
    no shutdown
    vni 10010
        vlan 10
    vni 10011
        vlan 11
    vni 100001
        routing
        vrf VRF1
! L3 VNI with VRF attached

vsx
! The same virtual-mac and VSX system-mac are recommended for a VSX logical VTEP pair
system-mac 02:00:00:00:01:00
inter-switch-link lag 256
role primary
keepalive peer 192.168.0.1 source 192.168.0.0 vrf KA
!
!
router ospf 1
    router-id 192.168.1.3
    area 0.0.0.0
router bgp 65001
    bgp router-id 192.168.1.3
    neighbor spine-RR peer-group
    neighbor spine-RR remote-as 65001
    neighbor spine-RR description Spine and RR peer-group
    neighbor spine-RR update-source loopback 0
    neighbor 192.168.1.1 peer-group spine-RR
    neighbor 192.168.1.2 peer-group spine-RR
    address-family l2vpn evpn
        neighbor spine-RR send-community extended
        neighbor 192.168.1.1 activate
        neighbor 192.168.1.2 activate
! Peer to EVPN RRs
exit-address-family
!
    vrf VRF1
        address-family ipv4 unicast
            redistribute connected
        exit-address-family

```



```

        address-family ipv6 unicast
            redistribute connected
        exit-address-family
! Used to advertise EVPN type 5 ip prefix routes
!
https-server vrf mgmt

```

The virtual-mac must be added to symmetric IRB Distributed L3 Gateways deployments. To help understand, a packet capture of the virtual-mac that is changed in the inner destination MAC field is shown below.

All remote VTEPs will send L3 VXLAN traffic with inner destination MAC changed to 02:00:00:00:01:00 as Leaf1A and Leaf1B are logically 1 VTEP and should have the same inner destination MAC.

```

- 28 2.1497... 10.1.12.10      10.1.10.100      ICMP 186 Echo (ping) reply  id=0x5b15, seq=1/256, ttl
+ Frame 28: 186 bytes on wire (1488 bits), 186 bytes captured (1488 bits) on interface \Device\NPF_{BC
+ Ethernet II, Src: HewlettP_fc:5c:00 (54:80:28:fc:5c:00), Dst: HewlettP_1b:13:70 (3c:4a:92:1b:13:70)
+ Internet Protocol Version 4, Src: 192.168.3.200, Dst: 192.168.3.251
+ Generic Routing Encapsulation (ERSPAN)
+ Encapsulated Remote Switch Packet Analysis Type I
+ Ethernet II, Src: ArubaHe_93:ca:40 (88:3a:30:93:ca:40), Dst: HewlettP_fd:f3:00 (54:80:28:fd:f3:00)
+ Internet Protocol Version 4, Src: 192.168.11.5, Dst: 192.168.11.3
+ User Datagram Protocol, Src Port: 36638, Dst Port: 4789
+ Virtual eXtensible Local Area Network
+ Flags: 0x0800, VXLAN Network ID (VNI)
+ Group Policy ID: 0
+ VXLAN Network Identifier (VNI): 100001
+ Reserved: 0
+ Ethernet II, Src: HewlettP_fc:ac:00 (54:80:28:fc:ac:00), Dst: 02:00:00:00:01:00 (02:00:00:00:01:00)
+ Destination: 02:00:00:00:01:00 (02:00:00:00:01:00)
+ Source: HewlettP_fc:ac:00 (54:80:28:fc:ac:00)
+ Type: IPv4 (0x0800)
+ Internet Protocol Version 4, Src: 10.1.12.10, Dst: 10.1.10.100
+ Internet Control Message Protocol

```

Sample Leaf1B Configuration

```

hostname Leaf1B
user admin group administrators password ciphertext AQBap!snip
vrf KA
vrf VRF1
    rd 65001:1
    route-target export 65001:1 evpn
    route-target import 65001:1 evpn
!
!
ssh server vrf mgmt
vlan 1
vlan 2
    vsx-sync
vlan 10
    vsx-sync
vlan 11
    vsx-sync
virtual-mac 02:00:00:00:01:00
! A unique virtual-mac is required on all symmetric IRB distributed L3 gateway VTEPs.
! It should be the same for a VSX logical VTEP pair, different virtual-macs should be used
! for other VSX VTEP pairs
! This is the inner D-MAC/egress VTEP router MAC

evpn
    vlan 10
        rd auto
        route-target export auto
        route-target import auto
        redistribute host-route
    vlan 11
        rd auto
        route-target export auto
        route-target import auto
        redistribute host-route

interface mgmt
    no shutdown
    ip static 10.136.40.238/24
    default-gateway 10.136.40.6
    nameserver 10.136.40.6

! Southbound server facing ports
interface lag 1 multi-chassis
    no shutdown

```

```

    description Server
    no routing
    vlan trunk native 1
    vlan trunk allowed 10-11
    lacp mode active
interface 1/1/1
    no shutdown
    lag 1

! Northbound uplink ports
interface 1/1/23
    no shutdown
    mtu 9198
    ip mtu 9198
    ip address 192.168.3.3/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point
interface 1/1/24
    no shutdown
    mtu 9198
    ip mtu 9198
    ip address 192.168.3.11/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point

! VSX ports
interface 1/1/41
    no shutdown
    vrf attach KA
    description VSX KeepAlive
    ip address 192.168.0.1/31
interface lag 256
    vsx-sync vlans
    no shutdown
    description ISL
    no routing
    vlan trunk native 1 tag
    vlan trunk allowed all
    lacp mode active
interface 1/1/49
    no shutdown
    mtu 9148
    description VSX ISL link
    lag 256
interface 1/1/50
    no shutdown
    mtu 9148
    description VSX ISL link
    lag 256

! Unique Lo0 on each VTEP
interface loopback 0
    ip address 192.168.1.4/32
    ip ospf 1 area 0.0.0.0

! Shared anycast Lo1 between a VSX logical VTEP pair
interface loopback 1
    ip address 192.168.11.3/32
    ip ospf 1 area 0.0.0.0

! Transit VLAN between VSX nodes for IGP continuity
interface vlan 2
    ip mtu 9198
    ip address 192.168.3.201/31
    ip ospf 1 area 0.0.0.0
    ip ospf cost 50
    ip ospf network point-to-point

! In 10.6, SVI IP (and/or IPv6) address must be different from the anycast Active Gateway IP (and/or IPv6) address
! this provides easier troubleshooting while sourcing traffic from the SVI IP (and/or IPv6) that is not
! common to all VTEPs
! Configuration simplification with same IP address for SVI and Active Gateway is an
! enhancement for a future release.
interface vlan 10
    vsx-sync active-gateways
    vrf attach VRF1
    ip address 10.1.10.3/24
    ipv6 address 1010::3/64
    active-gateway ip mac 12:00:00:00:01:00
    active-gateway ip 10.1.10.1
    active-gateway ipv6 mac 12:00:00:00:01:00
    active-gateway ipv6 fe80::10
    ip helper-address 192.168.3.242 vrf default
interface vlan 11
    vsx-sync active-gateways
    vrf attach VRF1

```

```

ip address 10.1.11.3/24
ipv6 address 1011::3/64
active-gateway ip mac 12:00:00:00:01:00
active-gateway 10.1.11.1
active-gateway ipv6 mac 12:00:00:00:01:00
active-gateway fe80::11
interface vxlan 1
source ip 192.168.11.3
vxlan-counters aggregate
no shutdown
vni 10010
    vlan 10
vni 10011
    vlan 11
vni 10020
    vlan 20
vni 100001
    routing
    vrf VRF1
! L3 VNI with VRF attached

vsx
! The same virtual-mac and VSX system-mac are recommended for a VSX logical VTEP pair
system-mac 02:00:00:00:01:00
inter-switch-link lag 256
role secondary
keepalive peer 192.168.0.0 source 192.168.0.1 vrf KA
!
!
router ospf 1
router-id 192.168.1.4
area 0.0.0.0
router bgp 65001
bgp router-id 192.168.1.4
neighbor spine-RR peer-group
neighbor spine-RR remote-as 65001
neighbor spine-RR description Spine and RR peer-group
neighbor spine-RR update-source loopback 0
neighbor 192.168.1.1 peer-group spine-RR
neighbor 192.168.1.2 peer-group spine-RR
address-family l2vpn evpn
    neighbor spine-RR send-community extended
    neighbor 192.168.1.1 activate
    neighbor 192.168.1.2 activate
! Peer to EVPN RRs
exit-address-family
!
vrf VRF1
address-family ipv4 unicast
    redistribute connected
exit-address-family
address-family ipv6 unicast
    redistribute connected
exit-address-family
! Used to advertise EVPN type 5 ip prefix routes
!
https-server vrf mgmt

```

Sample Leaf2A Configuration

```

hostname Leaf2A
user admin group administrators password ciphertext AQBap!snip
vrf KA
vrf VRF1
    rd 65001:2
    route-target export 65001:1 evpn
    route-target import 65001:1 evpn
!
!
ssh server vrf mgmt
vlan 1
vlan 2
    vsx-sync
vlan 10
    vsx-sync
vlan 12
    vsx-sync
virtual-mac 02:00:00:00:02:00
! A unique virtual-mac is required on all symmetric IRB distributed L3 gateway VTEPs.
! It should be the same for a VSX logical VTEP pair, different virtual-macs should be used
! for other VSX VTEP pairs
! This is the inner D-MAC/egress VTEP router MAC

evpn
    vlan 10

```

```

        rd auto
        route-target export auto
        route-target import auto
        redistribute host-route
    vlan 12
        rd auto
        route-target export auto
        route-target import auto
        redistribute host-route
! redistribute host-route is used to advertise host routes (/32 for IPv4, /128 for IPv6) to remote VTEPs
! e.g. 10.1.10.100/32 and/or IPv6 address 1010::100/128 is connected to Leaf1A/Leaf1B
! 10.1.12.10/32 and/or IPv6 address 1012::10/128 is connected to Leaf2A/Leaf2B
! This is required as the same subnet is spread across multiple VTEPs
! Without host routes, a remote VTEP would not be aware of the correct destination VTEP
! to send traffic
!

interface mgmt
    no shutdown
    ip static 10.136.40.233/24
    default-gateway 10.136.40.6

! Southbound server facing ports
interface lag 37 multi-chassis
    no shutdown
    description Server
    no routing
    vlan trunk native 1
    vlan trunk allowed 10
    lacp mode active
interface lag 39 multi-chassis
    no shutdown
    description Server
    no routing
    vlan trunk native 1
    vlan trunk allowed 12
    lacp mode active
interface 1/1/37
    no shutdown
    lag 37
interface 1/1/39
    no shutdown
    lag 39

! Northbound uplink ports
interface 1/1/23
    no shutdown
    mtu 9198
    ip mtu 9198
    ip address 192.168.3.5/31
    ip ospf 1 area 0.0.0.0
    no ip ospf passive
    ip ospf network point-to-point
interface 1/1/24
    no shutdown
    mtu 9198
    ip mtu 9198
    ip address 192.168.3.13/31
    ip ospf 1 area 0.0.0.0
    no ip ospf passive
    ip ospf network point-to-point

! VSX ports
interface 1/1/25
    no shutdown
    vrf attach KA
    description VSX KeepAlive
    ip address 192.168.0.2/31
interface lag 256
    vsx-sync vlans
    no shutdown
    description ISL LAG
    no routing
    vlan trunk native 1 tag
    vlan trunk allowed all
    lacp mode active
interface 1/1/55
    no shutdown
    mtu 9198
    description VSX ISL link
    lag 256
interface 1/1/56
    no shutdown
    mtu 9198
    description VSX ISL link
    lag 256

```

```

! Unique Lo0 on each VTEP
interface loopback 0
 ip address 192.168.1.5/32
 ip ospf 1 area 0.0.0.0

! Shared anycast Lo1 between a VSX logical VTEP pair
interface loopback 1
 ip address 192.168.11.5/32
 ip ospf 1 area 0.0.0.0

! Transit VLAN between VSX nodes for IGP continuity
interface vlan 2
 ip mtu 9198
 ip address 192.168.3.202/31
 ip ospf 1 area 0.0.0.0
 no ip ospf passive
 ip ospf cost 50
 ip ospf network point-to-point

! In 10.6, SVI IP (and/or IPv6) address must be different from the anycast Active Gateway IP(and/or IPv6)address
! this provides easier troubleshooting while sourcing traffic from the SVI IP (and/or IPv6) that is not
! common to all VTEPs
! Configuration simplification with same IP address for SVI and Active Gateway is an
! enhancement for a future release.
interface vlan 10
 vsx-sync active-gateways
 vrf attach VRF1
 ip address 10.1.10.4/24
 ipv6 address 1010::4/64
 active-gateway ip mac 12:00:00:00:01:00
 active-gateway ip 10.1.10.1
 active-gateway ipv6 mac 12:00:00:00:01:00
 active-gateway ipv6 fe80::10
interface vlan 12
 vsx-sync active-gateways
 vrf attach VRF1
 ip address 10.1.12.2/24
 ipv6 address 1012::2/64
 active-gateway ip mac 12:00:00:00:01:00
 active-gateway ip 10.1.12.1
 active-gateway ipv6 mac 12:00:00:00:01:00
 active-gateway ipv6 fe80::12

interface vxlan 1
 source ip 192.168.11.5
 no shutdown
 vni 10010
   vlan 10
 vni 10012
   vlan 12
 !
 vni 100001
   routing
   vrf VRF1
! L3 VNI with VRF attached

vsx
! the same virtual-mac and VSX system-mac are recommended for a VSX logical VTEP pair
 system-mac 02:00:00:00:02:00
 inter-switch-link lag 256
 role primary
 keepalive peer 192.168.0.3 source 192.168.0.2 vrf KA
!
router ospf 1
 router-id 192.168.1.5
 area 0.0.0.0
router bgp 65001
 bgp router-id 192.168.1.5
 neighbor spine-RR peer-group
 neighbor spine-RR remote-as 65001
 neighbor spine-RR description Spine and RR peer-group
 neighbor spine-RR update-source loopback 0
 neighbor 192.168.1.1 peer-group spine-RR
 neighbor 192.168.1.2 peer-group spine-RR
 address-family l2vpn evpn
   neighbor spine-RR send-community extended
   neighbor 192.168.1.1 activate
   neighbor 192.168.1.2 activate
! Peer to EVPN RRs
 exit-address-family
!
 vrf VRF1
   address-family ipv4 unicast
   exit-address-family
   address-family ipv6 unicast

```

```

        redistribute connected
        exit-address-family
! Used to advertise EVPN type 5 ip prefix routes
!
https-server vrf mgmt

```

Sample Leaf2B Configuration

```

hostname Leaf2B
user admin group administrators password ciphertext AQBap!snip
vrf KA
vrf VRF1
    rd 65001:2
    route-target export 65001:1 evpn
    route-target import 65001:1 evpn
!
!
ssh server vrf mgmt
vlan 1
vlan 2
    vsx-sync
vlan 10
    vsx-sync
vlan 12
    vsx-sync

virtual-mac 02:00:00:00:02:00
! A unique virtual-mac is required on all symmetric IRB distributed L3 gateway VTEPs.
! It should be the same for a VSX logical VTEP pair, different virtual-macs should be used
! for other VSX VTEP pairs.
! This is the inner D-MAC/egress VTEP router MAC.

evpn
    vlan 10
        rd auto
        route-target export auto
        route-target import auto
        redistribute host-route
    vlan 12
        rd auto
        route-target export auto
        route-target import auto
        redistribute host-route
! redistribute host-route is used to advertise host routes (/32 for IPv4, /128 for IPv6) to remote VTEPs
! e.g. 10.1.10.100/32 or an IPv6 address 1010::100/128 is connected to Leaf1A/Leaf1B
! 10.1.12.10/32 or IPv6 address 1012::100/128 is connected to Leaf2A/Leaf2B
! This is required as the same subnet is spread across multiple VTEPs
! Without host routes, a remote VTEP would not be aware of the correct destination VTEP
! to send traffic
!

interface mgmt
    no shutdown
    ip static 10.136.40.234/24
    default-gateway 10.136.40.6

! Southbound server facing ports
interface lag 37 multi-chassis
    no shutdown
    description Server
    no routing
    vlan trunk native 1
    vlan trunk allowed 10
    lacp mode active
interface lag 39 multi-chassis
    no shutdown
    description Server
    no routing
    vlan trunk native 1
    vlan trunk allowed 12
    lacp mode active
interface 1/1/37
    no shutdown
    lag 37
interface 1/1/39
    no shutdown
    lag 39

! Northbound uplink ports
interface 1/1/23
    no shutdown
    mtu 9198
    ip mtu 9198
    ip address 192.168.3.7/31
    ip ospf 1 area 0.0.0.0

```

```

    ip ospf network point-to-point
interface 1/1/24
    no shutdown
    mtu 9198
    ip mtu 9198
    ip address 192.168.3.15/31
    ip ospf 1 area 0.0.0.0
    ip ospf network point-to-point

! VSX ports
interface 1/1/25
    no shutdown
    vrf attach KA
    description keepalive
    ip address 192.168.0.3/31
interface lag 256
    vsx-sync vlans
    no shutdown
    description ISL LAG
    no routing
    vlan trunk native 1 tag
    vlan trunk allowed all
    lacp mode active
interface 1/1/55
    no shutdown
    mtu 9198
    description VSX ISL link
    lag 256
interface 1/1/56
    no shutdown
    mtu 9198
    description VSX ISL link
    lag 256

! Unique Lo0 on each VTEP
interface loopback 0
    ip address 192.168.1.6/32
    ip ospf 1 area 0.0.0.0

! Shared anycast Lo1 between a VSX logical VTEP pair
interface loopback 1
    ip address 192.168.11.5/32
    ip ospf 1 area 0.0.0.0

! Transit VLAN between VSX nodes for IGP continuity
interface vlan 2
    ip mtu 9198
    ip address 192.168.3.203/31
    ip ospf 1 area 0.0.0.0
    ip ospf cost 50
    ip ospf network point-to-point

! In 10.6, SVI IP (and/or IPv6) address must be different from the anycast Active Gateway IP(and/or IPv6)address
! this provides easier troubleshooting while sourcing traffic from the SVI IP (and/or IPv6) that is not
! common to all VTEPs
! Configuration simplification with same IP address for SVI and Active Gateway is an
! enhancement for a future release.
interface vlan 10
    vsx-sync active-gateways
    vrf attach VRF1
    ip address 10.1.10.5/24
    ipv6 address 1010::5/64
    active-gateway ip mac 12:00:00:00:01:00
    active-gateway ip 10.1.10.1
    active-gateway ipv6 mac 12:00:00:00:01:00
    active-gateway ipv6 fe80::10
interface vlan 12
    vsx-sync active-gateways
    vrf attach VRF1
    ip address 10.1.12.3/24
    ipv6 address 1012::3/64
    active-gateway ip mac 12:00:00:00:01:00
    active-gateway ip 10.1.12.1
    active-gateway ipv6 mac 12:00:00:00:01:00
    active-gateway ipv6 fe80::12
interface vxlan 1
    source ip 192.168.11.5
    no shutdown
    vni 10010
        vlan 10
    vni 10012
        vlan 12
    !
    vni 100001
        routing
        vrf VRF1

```

```

! L3 VNI with VRF attached

vsx
! the same virtual-mac and VSX system-mac are recommended for a VSX logical VTEP pair
  system-mac 02:00:00:00:02:00
  inter-switch-link lag 256
  role secondary
  keepalive peer 192.168.0.2 source 192.168.0.3 vrf KA
!
router ospf 1
  router-id 192.168.1.6
  area 0.0.0.0
router bgp 65001
  bgp router-id 192.168.1.6
  neighbor spine-RR peer-group
  neighbor spine-RR remote-as 65001
  neighbor spine-RR description Spine and RR peer-group
  neighbor spine-RR update-source loopback 0
  neighbor 192.168.1.1 peer-group spine-RR
  neighbor 192.168.1.2 peer-group spine-RR
  address-family l2vpn evpn
    neighbor spine-RR send-community extended
    neighbor 192.168.1.1 activate
    neighbor 192.168.1.2 activate
! Peer to EVPN RRs
  exit-address-family
!
  vrf VRF1
    address-family ipv4 unicast
      redistribute connected
    exit-address-family
    address-family ipv6 unicast
      redistribute connected
    exit-address-family
! Used to advertise EVPN type 5 ip prefix routes
!
https-server vrf mgmt

```

Sample DHCP relay with Ubuntu DHCP Server Configuration

Update and install the DHCP server:

```

sudo apt get update
sudo apt-get install isc-dhcp-server -y

```

Editing dhcpd.conf

```

sudo vi /etc/dhcp/dhcpd.conf
option domain-name "local.*****.net";
option domain-name-servers 8.8.8.8, 8.8.8.4;
authoritative;

subnet 192.168.3.240 netmask 255.255.255.248 {}
# isc-dhcp-server requires local subnet on DHCP server to be configured in order to start

# Here are 2 example DHCP scopes
# Routers and dhcp-server-identifier would match Active Gateway IP(and/or IPv6) on the SVIs
subnet 10.1.10.0 netmask 255.255.255.0 {
  range 10.1.10.10 10.1.10.40;
  option routers 10.1.10.1;
  option dhcp-server-identifier 10.1.10.1;
}

subnet 10.1.11.0 netmask 255.255.255.0 {
  range 10.1.11.10 10.1.11.40;
  option routers 10.1.11.1;
  option dhcp-server-identifier 10.1.11.1;
}
# save and exit editor

```

VRFs with overlapping subnets

```

sudo vi /etc/dhcp/dhcpd.conf
option domain-name "local.*****.net";
option domain-name-servers 8.8.8.8, 8.8.8.4;

```



```

authoritative;

subnet 192.168.3.240 netmask 255.255.255.248 {}
# isc-dhcp-server requires local subnet on DHCP server to be configured in order to start

# The AOS-CX DHCP relay VTEP will send VRF name as part of option 82, sub-option 151
class "VRF1"
{
    match if substring (option agent.unknown-151,1,100) = "VRF1";
}
class "VRF2"
{
    match if substring (option agent.unknown-151,1,100) = "VRF2";
}
subnet 10.1.10.0 netmask 255.255.255.0 {
    option routers 10.1.10.1;
    pool {
        allow members of "VRF1";
        range 10.1.10.10 10.1.10.40;
    }
    pool {
        allow members of "VRF2";
        range 10.1.10.41 10.1.10.60;
    }
}
# save and exit editor

```

Start and check DHCP server status

```

sudo systemctl start isc-dhcp-server.service
sudo systemctl status isc-dhcp-server.service

```

Additional configuration required on a VTEP for DHCP relay, ensure network connectivity works between Lo0 and DHCP server through VRF default

```

interface loopback 0
    ip address 192.168.1.3/32
!
ip source-interface dhcp_relay interface loopback0
dhcp-relay option 82 source-interface
! The above commands ensure that VTEP DHCP relay communication uses Lo0 to DHCP server
! and DHCP-relay option 82, sub-option 5, 11 and 151 are sent to the DHCP server using Lo0 IP
!
interface vlan10
    vrf attach VRF1
    ip address 10.1.10.2/24
    ip helper-address 192.168.3.242 vrf default
! Point to DHCP server in VRF default
    active-gateway ip mac 12:00:00:00:01:00
    active-gateway ip 10.1.10.1
!
interface vlan11
    vrf attach VRF2
    ip address 10.1.11.2/24
    ip helper-address 192.168.3.242 vrf default
! Point to DHCP server in VRF default
    active-gateway ip mac 12:00:00:00:01:00
    active-gateway ip 10.1.11.1

```

A packet capture of the interaction between the DHCP relay VTEP and DHCP server is shown below with option 82 and sub-options 5, 11 and 151 with VRF1 in the contents of VRF name field.

24	4.4932...	0.0.0.0	255.255.255.2...	DHCP	342	DHCP Discover	- Transaction ID 0x1c883133
25	4.4934...	192.168.1.3	192.168.3.242	DHCP	351	DHCP Discover	- Transaction ID 0x1c883133
37	5.4944...	192.168.3.242	192.168.1.3	DHCP	368	DHCP Offer	- Transaction ID 0x1c883133
38	5.4969...	0.0.0.0	255.255.255.2...	DHCP	342	DHCP Request	- Transaction ID 0x1c883133
39	5.4971...	192.168.1.3	192.168.3.242	DHCP	357	DHCP Request	- Transaction ID 0x1c883133
40	5.5084...	192.168.3.242	192.168.1.3	DHCP	368	DHCP ACK	- Transaction ID 0x1c883133
41	5.5087...	192.168.3.242	192.168.1.3	DHCP	368	DHCP ACK	- Transaction ID 0x1c883133

```

- Parameter Request List Item: (1) Subnet Mask
- Parameter Request List Item: (28) Broadcast Address
- Parameter Request List Item: (2) Time Offset
- Parameter Request List Item: (3) Router
- Parameter Request List Item: (15) Domain Name
- Parameter Request List Item: (6) Domain Name Server
- Parameter Request List Item: (119) Domain Search
- Parameter Request List Item: (12) Host Name
- Parameter Request List Item: (44) NetBIOS over TCP/IP Name Server
- Parameter Request List Item: (47) NetBIOS over TCP/IP Scope
- Parameter Request List Item: (26) Interface MTU
- Parameter Request List Item: (121) Classless Static Route
- Parameter Request List Item: (42) Network Time Protocol Servers

```

Option: (82) Agent Information Option

```

- Length: 35
- Option 82 Suboption: (1) Agent Circuit ID
  - Length: 4
  - Agent Circuit ID: 0000000a
- Option 82 Suboption: (2) Agent Remote ID
  - Length: 6
  - Agent Remote ID: 120000000100
- Option 82 Suboption: (5) Link selection (10.1.10.2)
  - Length: 4
  - Link selection: 10.1.10.2
- Option 82 Suboption: (11) Server ID Override (10.1.10.2)
  - Length: 4
  - Server ID Override: 10.1.10.2

```

Option 82 Suboption: (151) VRF name/VPN ID

```

- Length: 5
- VRF name:

```

```

- [Expert Info (Warning/Undecoded): Trailing stray characters]

```

```

f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
30 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
10 00 00 00 00 00 00 63 82 53 63 35 01 03 36 04 c0 .....c·Sc5··6·
20 a8 03 f2 32 04 0a 01 0a 64 0c 05 76 75 62 75 32 ...2·... d·vubu2
30 37 0d 01 1c 02 03 0f 06 77 0c 2c 2f 1a 79 2a 52 7·... w·,/.y*R
40 23 01 04 00 00 00 0a 02 06 12 00 00 00 01 00 05 #·...
50 04 0a 01 0a 02 0b 04 0a 01 0a 02 97 05 00 56 52 .....·VR
50 46 31 98 00 ff F1...

```

Sample DHCP relay with Windows DHCP Server Configuration

The Windows Server 2016 DHCP server requires VTEP loopbacks to be added as scopes that are excluded from DHCP allocation. For example, Scope 10.1.2.0 and 10.1.1.0 are IP ranges assigned to VTEP loopbacks.

If VTEP loopback scopes are not added, DHCP clients would not be able to correctly grab DHCP IPs.

DHCP			
ds-dhcp			
IPv4			
Scope [10.1.2.0] Loopbacks Pod2			
Address Pool			
Address Leases			
Reservations			
Scope Options			
Policies			

Start IP Address	End IP Address	Description
10.1.2.1	10.1.2.8	Address range for distribution
10.1.2.1	10.1.2.8	IP Addresses excluded from distribution

DHCP	
ds-dhcp	
IPv4	
Scope [10.1.2.0] Loopbacks Pod2	
Address Pool	
Address Leases	
Reservations	
Scope Options	
Policies	
Scope [10.1.1.0] Loopbacks Pod1	

	Scope Options
Scope options are additional configuration parameters that a DHCP server can assign to DHCP clients. For example, you can configure a DHCP server to assign a specific IP address to a specific client.	
Scope options override server options.	
To set the scope options, on the Action menu, click Configure Options.	
For more information about scope options, see online Help.	

Windows Server 2016 DHCP server does not require additional `dhcp-server-identifier` option to be added when compared to Ubuntu DHCP server.

DHCP			
ds-dhcp			
IPv4			
Scope [10.1.2.0] Loopbacks Pod2			
Scope [10.1.1.0] Loopbacks Pod1			
Scope [10.201.0.0] Pod1-VLAN1001			
Address Pool			
Address Leases			
Reservations			
Scope Options			
Policies			

Start IP Address	End IP Address	Description
10.201.0.201	10.201.0.250	Address range for distribution

DHCP			
ds-dhcp			
IPv4			
Scope [10.1.2.0] Loopbacks Pod2			
Scope [10.1.1.0] Loopbacks Pod1			
Scope [10.201.0.0] Pod1-VLAN1001			
Address Pool			
Address Leases			
Reservations			
Scope Options			
Policies			

Option Name	Vendor	Value	Policy Name
003 Router	Standard	10.201.0.1	None
006 DNS Servers	Standard	10.108.45.19	None

EVPN VSX support

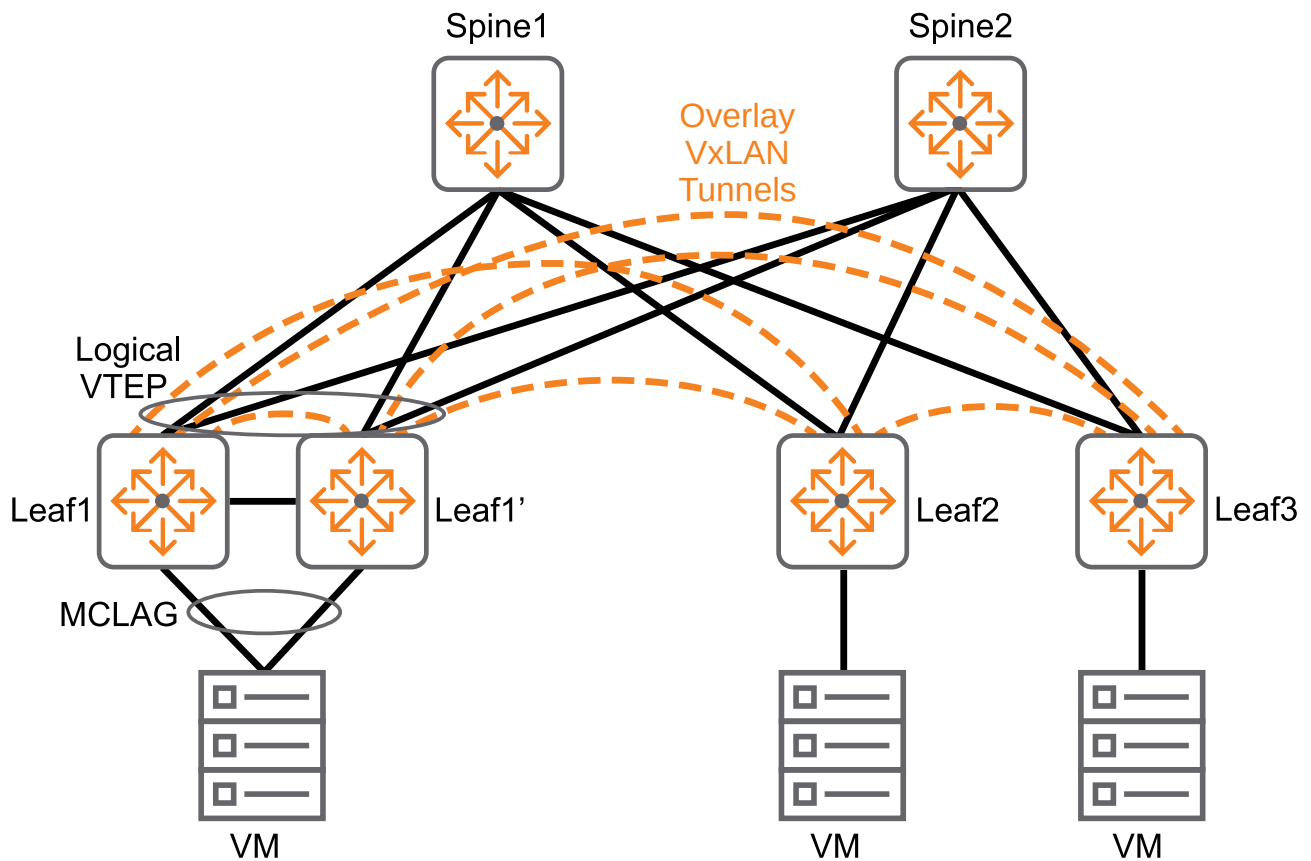


NOTE: Not applicable to 6300 switch series, since 6300 supports VSF virtualization technology instead of VSX.

VSX is a virtualization technology that presents the VSX pair as a single device to the layer 2 clients and as independent devices at layer 3. The VSX pair provides an MLAG link towards the clients. The clients see the VSX-pair as a single switch, providing a redundant L2 connection to the clients. VSX acts in active-active mode which also provides load-balancing to the connected clients.

EVPN is supported with VSX. The two VSX nodes act as independent BGP routing entities to the other VTEPs or spines for control packets. However, in the datapath, both of them act as a single logical VTEP. This is achieved by using different IP addresses for establishing the BGP session and using a common IP as next-hop to represent the VTEP.

EVPN VSX support: Logical VTEP



Both the VSX pairs are configured with a common Logical VTEP. The configuration has to be done on a loopback interface on both the VSX pairs. The underlay routing protocol (OSPF or even BGP) distributes this logical VTEP IP address to all other VTEPs.

The Overlay routes are redistributed with the next-hop as the Logical VTEP IP address. The Overlay packets reaching the Ingress VTEP get VXLAN encapsulated with the destination as the next-hop IP (common Logical VTEP if the destination is the VSX pair) and sent to one of the Spines.

The Spine does ECMP on the VXLAN packet and sends it to either Leaf1 or Leaf1' since both VSX switches have connectivity to the common Logical VTEP.

Configuration recommendations for EVPN VSX support

- Split recovery is enabled by default. While using EVPN and VSX together, this mode must be kept disabled to ensure that traffic always flows through the primary when ISL and Keepalive are both down.
- All the single homed hosts must be connected to the primary VSX Node. This is to ensure that traffic to single homed hosts are not affected when ISL/Secondary goes down.
- The same Route Distinguisher (RD) are configured in both the VSX peers. The RD must be unique across all the leafs except between VSX Peers.
- Even in case of EBGp peering, both the VSX peers must be in the same AS.
- The same VTEP IP must be configured on both the VSX peers.
- The BGP source IP must be configured in both the VSX peers and it must be different from the logical VTEP. This is to ensure that logical VTEP is not used as the source IP for the BGP session establishment with the neighbor.

- Routing session between VSX peers is recommended in case of upstream connectivity failure.
- `virtual-mac` is required for VSX distributed L3 gateways. The `virtual-mac` must be the same as the `VSX system-mac` for a logical VTEP. The `virtual-mac` must be unique on each VTEP.



NOTE: VSX active-sync is not supported for the VXLAN interface. Ensure that the configurations are manually configured between the VSX switches.

Overlay VLANs must be allowed on the ISL for MAC/ARP sync so both devices can forward traffic.

`vsx-sync` is not supported in EVPN feature-group.

For more details on VSX configurations, see the *Virtual Switching Extension (VSX) Guide*.

Sample configuration for iBGP VSX EVPN

The following sample configuration solution shows the relevant EVPN/VSX configurations on the primary and secondary VSX switches.

Spine configuration

```
interface loopback 1
    ip address 3.3.3.3/32
    ip ospf 1 area 0.0.0.0
router bgp 1
    bgp router-id 3.3.3.3
    neighbor 2.2.2.2 remote-as 1
    neighbor 2.2.2.2 update-source 3.3.3.3
    neighbor 10.10.10.10 remote-as 1
    neighbor 10.10.10.10 update-source 3.3.3.3
    neighbor 20.20.20.20 remote-as 1
    neighbor 20.20.20.20 update-source 3.3.3.3
    address-family l2vpn evpn
        neighbor 2.2.2.2 activate
        neighbor 2.2.2.2 route-reflector-client
        neighbor 2.2.2.2 send-community extended
        neighbor 10.10.10.10 activate
        neighbor 10.10.10.10 route-reflector-client
        neighbor 10.10.10.10 send-community extended
        neighbor 20.20.20.20 activate
        neighbor 20.20.20.20 route-reflector-client
        neighbor 20.20.20.20 send-community extended
    exit-address-family
```

Leaf1 configuration (Primary VTEP)

```
vlan 1-2
evpn
    vlan 2
        rd 1.1.1.1:2
        route-target export 2:2
        route-target import 2:2
interface 1/1/17
    no shutdown
    no routing
    vlan trunk native 1 tag
    vlan trunk allowed all
interface loopback 1
```

```

description "vxlan source-ip"
  ip address 1.1.1.1/32
  ip ospf 1 area 0.0.0.0
interface loopback 2
  ip address 10.10.10.10/32
  ip ospf 1 area 0.0.0.0
interface vlan2
  ip address 10.1.1.5/24
  active-gateway ip mac 00:00:00:00:00:33
  active-gateway ip 10.1.1.10
vsx
  inter-switch-link 1/1/17
  role primary
  keepalive peer 20.1.1.2 source 20.1.1.1

interface vxlan 1
  source ip 1.1.1.1
  no shutdown
  vni 100
    vlan 2
  !
router bgp 1
  bgp router-id 10.10.10.10
  neighbor 3.3.3.3 remote-as 1
  neighbor 3.3.3.3 update-source 10.10.10.10
  address-family l2vpn evpn
    neighbor 3.3.3.3 activate
    neighbor 3.3.3.3 send-community extended
  exit-address-family

```

Leaf1' configuration (Secondary VTEP)

```

vlan 1-2
evpn
  vlan 2
    rd 1.1.1.1:2
    route-target export 2:2
    route-target import 2:2
interface 1/1/23
  no shutdown
  no routing
  vlan trunk native 1 tag
  vlan trunk allowed all
interface loopback 1
description "vxlan source-ip"
  ip address 1.1.1.1/32
  ip ospf 1 area 0.0.0.0
interface loopback 2
  ip address 20.20.20.20/32
  ip ospf 1 area 0.0.0.0
interface vlan2
  ip address 10.1.1.6/24
  active-gateway ip mac 00:00:00:00:00:33
  active-gateway ip 10.1.1.10
vsx
  inter-switch-link 1/1/23
  role secondary
  keepalive peer 20.1.1.1 source 20.1.1.2

```

```

interface vxlan 1
    source ip 1.1.1.1
    no shutdown
    vni 100
    vlan 2
!
router bgp 1
    bgp router-id 20.20.20.20
    neighbor 3.3.3.3 remote-as 1
    neighbor 3.3.3.3 update-source 20.20.20.20
    address-family l2vpn evpn
        neighbor 3.3.3.3 activate
        neighbor 3.3.3.3 send-community extended
    exit-address-family

```

Leaf2 configuration

```

vlan 1-2
evpn
    vlan 2
        rd 2.2.2.2:2
        route-target export 2:2
        route-target import 2:2
interface loopback 1
    ip address 2.2.2.2/32
    ip ospf 1 area 0.0.0.0
interface vlan2
    ip address 10.1.1.7/24
    active-gateway ip mac 00:00:00:00:00:33
    active-gateway ip 10.1.1.10
interface vxlan 1
    source ip 2.2.2.2
    no shutdown
    vni 100
    vlan 2
!
router bgp 1
    neighbor 3.3.3.3 remote-as 1
    neighbor 3.3.3.3 update-source 2.2.2.2
    address-family l2vpn evpn
        neighbor 3.3.3.3 activate
        neighbor 3.3.3.3 send-community extended
    exit-address-family

```

VSX failure scenarios

Failure scenarios	Result with split recovery off	Result with split recovery on (default)
VTEP uplink failure	No impact to logical VTEP. Loopback connectivity between VTEPs will reroute through other available links using OSPF/BGP.	No impact to logical VTEP. Loopback connectivity between VTEPs will reroute through other available links using OSPF/BGP.
VTEP downlink failure	No impact to logical VTEP. Only VSX LAG to server is impacted.	No impact to logical VTEP. Only VSX LAG to server is impacted.
a) VSX Keepalive down. ISL up. b) Then after sometime, ISL goes down.	No impact to logical VTEP (but loss of split detection). No impact to logical VTEP (but loss of split detection and ISL sync).	No impact to logical VTEP (but loss of split detection). No impact to logical VTEP (but loss of split detection and ISL sync).  NOTE: Without ISL ARP sync, in routing scenarios, this split condition may lead to traffic loss where ARP request originated from a VSX device and reply lands on the peer VSX device.
a) ISL down, VSX Keepalive up.	Secondary VSX node tears down VSX LAG member ports and brings down logical VTEP to ensure that VXLAN traffic is only sent to the primary VSX switch.	Secondary VSX node tears down VSX LAG member ports and brings down logical VTEP to ensure VXLAN traffic is only sent to the primary VSX switch.

Table Continued

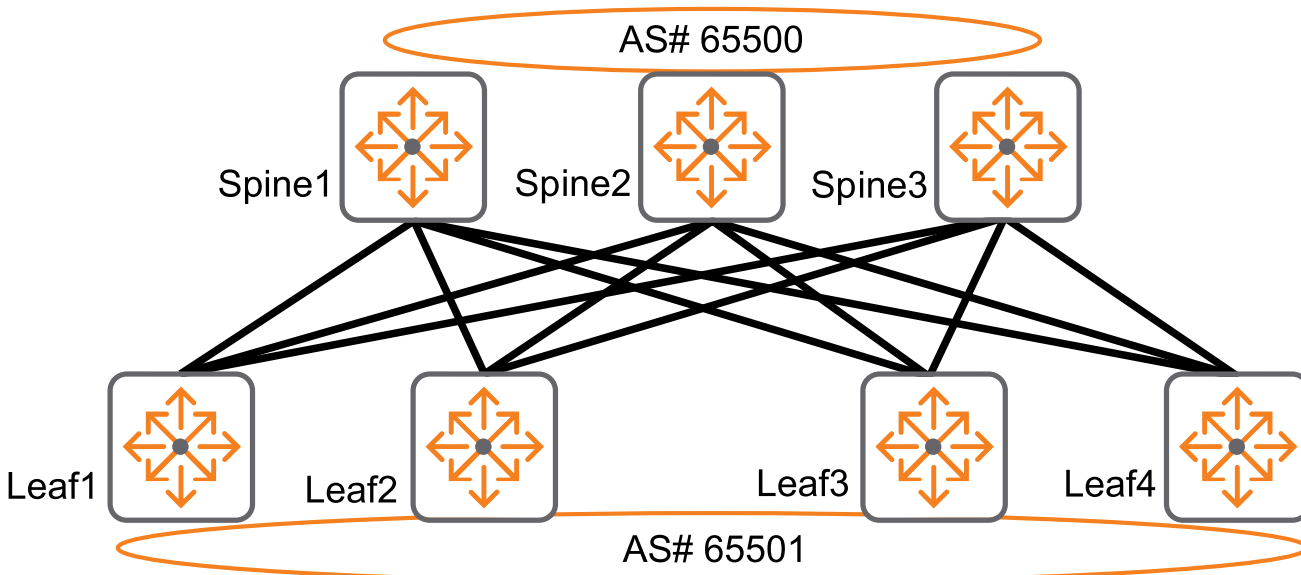
Failure scenarios	Result with split recovery off	Result with split recovery on (default)
a) ISL down, VSX Keepalive up. b) Then, after sometime, VSX Keepalive down as well.	Secondary VSX node tears down VSX LAG member ports and brings down logical VTEP to ensure VXLAN traffic is only sent to the primary VSX switch. Secondary VSX LAGs and logical VTEP stay down.	Secondary VSX node tears down VSX LAG member ports and brings down logical VTEP to ensure that VXLAN traffic is only sent to the primary VSX switch. Secondary VSX node restores VSX LAG member ports and brings up logical VTEP.  NOTE: Without ISL ARP sync, in routing scenarios, this split condition may lead to traffic loss where ARP request originated from a VSX device and reply lands on the peer VSX device.
a) ISL and keepalive are down b) Keepalive restore	Secondary VSX node tears down VSX LAG member ports and brings down logical VTEP(If it was UP earlier) to ensure that VXLAN traffic is only sent to primary VSX switch.	Secondary VSX node tears down VSX LAG member ports and brings down logical VTEP to ensure that VXLAN traffic is only sent to primary VSX switch.

For more details on VSX failure scenarios, see *Troubleshooting* in the *Virtual Switching Extension (VSX) Guide*.

eBGP support for EVPN

The BGP session for underlay and overlay can be either iBGP or eBGP. eBGP EVPN is used for Data Center use cases. Dual-AS and Multi-AS eBGP topologies are supported.

In a Dual-AS topology, all spines share a common AS number and all leafs share another common AS number. For example:



In a Dual-AS topology, since the peer VTEPs (leafs) receive the route from another leaf with the same AS number, the route would be rejected. To avoid this, use the following `allowas-in` command:

```
switch(config-bgp-l2vpn-evpn)# neighbor 1.1.1.1 allowas-in 1
```

A sample Dual-AS configuration is shown below.

Leaf 1 configuration

```
vlan 1-2
evpn
    vlan 2
        rd 5:5
        route-target export 1:1
        route-target import 1:1
interface 1/1/1
    no shutdown
    ip address 11.1.1.2/24
interface loopback 1
    ip address 2.2.2.2/32
interface vxlan 1
    source ip 2.2.2.2
    no shutdown
    vni 100
interface vlan 2
router bgp 65501
    neighbor 1.1.1.1 remote-as 65501
    neighbor 1.1.1.1 update-source 2.2.2.2
    neighbor 11.1.1.1 remote-as 65501
    address-family ipv4 unicast
        neighbor 11.1.1.1 activate
        neighbor 11.1.1.1 allowas-in 1
        network 2.2.2.2/32
    exit-address-family
    address-family l2vpn evpn
        neighbor 1.1.1.1 activate
        neighbor 1.1.1.1 allowas-in 1
```

```
neighbor 1.1.1.1 send-community extended
exit-address-family
```

Leaf 2 configuration

```
vlan 1-2
evpn
    vlan 2
        rd 5:5
        route-target export 1:1
        route-target import 1:1
interface 1/1/24
    no shutdown
    ip address 12.1.1.2/24
interface loopback 1
    ip address 3.3.3.3/32
interface vxlan 1
    source ip 3.3.3.3
    no shutdown
    vni 100
interface vlan 2
    !
router bgp 65501
    neighbor 1.1.1.1 remote-as 65501
    neighbor 1.1.1.1 update-source 3.3.3.3
    neighbor 12.1.1.1 remote-as 65501
    address-family ipv4 unicast
        neighbor 12.1.1.1 activate
        network 3.3.3.3/32
    exit-address-family
    address-family l2vpn evpn
        neighbor 1.1.1.1 activate
        neighbor 1.1.1.1 allowas-in 1
        neighbor 1.1.1.1 send-community extended
    exit-address-family
```

Spine configuration

```
vlan 1-2

interface 1/1/2
    no shutdown
    ip address 11.1.1.1/24
interface 1/1/5
    no shutdown
    ip address 12.1.1.1/24
interface loopback 1
    ip address 1.1.1.1/32
router bgp 65500
    neighbor 2.2.2.2 remote-as 65500
    neighbor 2.2.2.2 update-source 1.1.1.1
    neighbor 3.3.3.3 remote-as 65500
    neighbor 3.3.3.3 update-source 1.1.1.1
    neighbor 11.1.1.2 remote-as 65500
    neighbor 12.1.1.2 remote-as 65500
    address-family ipv4 unicast
        neighbor 11.1.1.2 activate
```

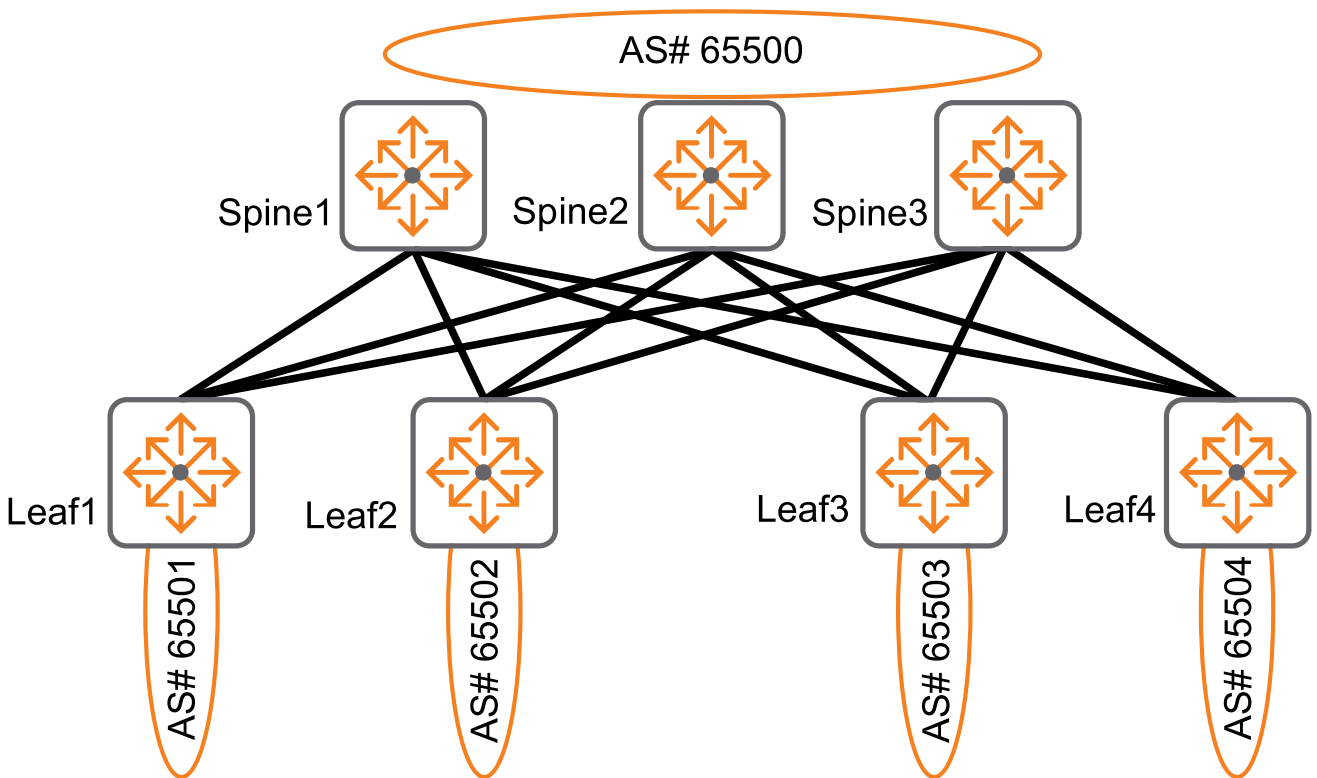
```

neighbor 12.1.1.2 activate
network 1.1.1.1/32
exit-address-family
address-family l2vpn evpn
neighbor 2.2.2.2 activate
neighbor 2.2.2.2 next-hop-unchanged
neighbor 2.2.2.2 send-community extended
neighbor 3.3.3.3 activate
neighbor 3.3.3.3 next-hop-unchanged
neighbor 3.3.3.3 send-community extended
exit-address-family

```

In a Multi-AS Topology, all spines share a common AS number and all leafs have different AS numbers.

Figure 1: Multi-AS Topology



A sample Multi-AS configuration is shown below.

Leaf 1 configuration

```

vlan 1-2
evpn
  vlan 2
    rd 5:5
    route-target export 1:1
    route-target import 1:1
interface 1/1/1
  no shutdown
  ip address 11.1.1.2/24
interface loopback 1
  ip address 2.2.2.2/32
interface vxlan 1

```

```

        source ip 2.2.2.2
        no shutdown
        vni 100
interface vlan 2
router bgp 65500
    neighbor 1.1.1.1 remote-as 65500
    neighbor 1.1.1.1 update-source 2.2.2.2
    neighbor 11.1.1.1 remote-as 65500
    address-family ipv4 unicast
        neighbor 11.1.1.1 activate
        network 2.2.2.2/32
    exit-address-family
    address-family l2vpn evpn
        neighbor 1.1.1.1 activate
        neighbor 1.1.1.1 send-community extended
    exit-address-family

```

Leaf 2 configuration

```

vlan 1-2
evpn
    vlan 2
        rd 5:5
        route-target export 1:1
        route-target import 1:1
interface 1/1/24
    no shutdown
    ip address 12.1.1.2/24
interface loopback 1
    ip address 3.3.3.3/32
interface vxlan 1
    source ip 3.3.3.3
    no shutdown
    vni 100
interface vlan 2
    !
router bgp 65500
    neighbor 1.1.1.1 remote-as 65500
    neighbor 1.1.1.1 update-source 3.3.3.3
    neighbor 12.1.1.1 remote-as 65500
    address-family ipv4 unicast
        neighbor 12.1.1.1 activate
        network 3.3.3.3/32
    exit-address-family
    address-family l2vpn evpn
        neighbor 1.1.1.1 activate
        neighbor 1.1.1.1 send-community extended
    exit-address-family

```

Spine configuration

```

vlan 1-2
interface 1/1/2
    no shutdown
    ip address 11.1.1.1/24
interface 1/1/5
    no shutdown

```

```

ip address 12.1.1.1/24
interface loopback 1
ip address 1.1.1.1/32
router bgp 65501
neighbor 2.2.2.2 remote-as 65501
neighbor 2.2.2.2 update-source 1.1.1.1
neighbor 3.3.3.3 remote-as 65502
neighbor 3.3.3.3 update-source 1.1.1.1
neighbor 11.1.1.2 remote-as 65503
neighbor 12.1.1.2 remote-as 65504
address-family ipv4 unicast
neighbor 11.1.1.2 activate
neighbor 12.1.1.2 activate
network 1.1.1.1/32
exit-address-family
address-family l2vpn evpn
neighbor 2.2.2.2 activate
neighbor 2.2.2.2 next-hop-unchanged
neighbor 2.2.2.2 send-community extended
neighbor 3.3.3.3 activate
neighbor 3.3.3.3 next-hop-unchanged
neighbor 3.3.3.3 send-community extended
exit-address-family

```

The tunnel endpoints are communicated as next-hops in the EVPN routes. By default, the next hop of a route is preserved when advertising the route to an iBGP peer, but is updated when advertising the route to an eBGP peer. Setting this to 'true' over-rides this behavior and preserves the next hop when routes are advertised to eBGP peer. To retain the next-hop (or tunnel endpoint info), an extra configuration is required as shown in the following command:

```
switch(config-bgp-l2vpn-evpn)# neighbor 2.2.2.2 next-hop-unchanged
```



NOTE: When VSX is used with eBGP, the VSX peers must be in same AS.

Route-targets must be manually configured for switches running eBGP since auto generated RT's (route-target auto) would lead to different export/import route-targets (Applicable for both VSX/non-VSX).

MAC mobility

MAC mobility refers to the movement of a user terminal from one VTEP to another. The source VTEP is unaware of the MAC move event. To notify other VTEPs of the change, the destination VTEP advertises a MAC advertisement route. The source VTEP withdraws the old route for the MAC address after receiving the new route. The MAC advertisement route has a sequence number that increases when the MAC address moves. The sequence number identifies the most recent move if the MAC address moves multiple times.

EVPN MAC dampening

Supported on the 6300, 6400, 8325, 8360, and 8400 Switch Series only

EVPN MAC dampening is used to detect duplicate MAC addresses in the network and provide a mechanism to protect against endless MAC moves.

As per RFC 7432, same MAC address is learned by different VTEPs in the same VLAN because two or more hosts were configured incorrectly with the same MAC address. The traffic from the hosts triggers continuous MAC moves among the VTEPs attached to the hosts. Every MAC mobility event for a given MAC address will contain a sequence number of MAC routers received in route type-2. Based on this sequence number of the

MAC received and the MAC that is learnt locally, a decision to further withdraw or send a route update is taken.

A VTEP receiving a MAC or IP advertisement route for a MAC address with a sequence number higher than the one it had previously advertised, withdraws its MAC or IP advertisement route.

A VTEP detecting a locally attached MAC address for which it had previously received a MAC or IP advertisement route, advertises the MAC address in a MAC or IP advertisement route. This advertisement route is tagged with a MAC Mobility extended community attribute with a sequence number one greater than the sequence number in the MAC Mobility extended community attribute of the received MAC or IP advertisement route.

For the switch to recognize such instances and prevent the sequence number to increment infinitely, the VTEP starts a timer when it detects a MAC mobility event involving a locally learnt MAC address. If the VTEP detects a MAC move before the timer expires, it concludes that a duplicate MAC situation has occurred. The VTEP alerts the operator through the event log and stops sending and processing any BGP MAC or IP advertisement routes for the MAC address until the right action is provided.



NOTE: In a VSX setup, MAC dampening does not work as expected. Because, the VSX primary dampens the MAC, but the VSX secondary will continue to advertise the dampened MAC.

Following is the command syntax to configure EVPN MAC dampening:

```
mac-move-detection count <mac-move-count> timer <mac-move-timer>
```

EVPN commands

active-gateway

Syntax

```
active-gateway {ip | ipv6} [<IP-ADDRESS>] [mac <MAC-ADDRESS>]
```

```
no active-gateway {ip | ipv6} [<IP-ADDRESS>] [mac]
```

Description

Configures an EVPN Anycast gateway that can be used on multiple VTEPs. The Active Gateway supports both IPv4 and IPv6 addresses. The Active Gateway MAC address used along with an IPv4 and IPv6 address must match on a given interface, for the EVPN Anycast Gateway solution to work as expected.

The `no` form of this command removes the active gateway for active-active routing.

Command context

```
config-if-vlan
```

Parameters

ip

Specifies the configuration of an IPv4 address.

ipv6

Specifies the configuration of an IPv6 address.

<IP-ADDRESS>

Specifies the IPv4 or IPv6 address.

- Syntax for IPv4: A.B.C.D
- Syntax for IPv6: A:B::C:D

<MAC-ADDRESS>

Specifies the Virtual MAC address. Syntax: xx:xx:xx:xx:xx:xx

Authority

Administrators or local user group members with execution rights for this command.

Usage

Before configuring active gateway, confirm that an IP address is on the SVI that is in the same subnet as the active gateway IP you are trying to configure. If an active gateway IP does not have an SVI IP with the same subnet, the CLI allows the configuration, but the active gateway IP will not be programmed in the kernel, resulting the active gateway to be unreachable.

Active forwarding cannot be configured when ICMP redirect is enabled. Enter the `no ip icmp redirect` command for disabling ICMP redirect.

It is highly recommended that you use an IPv6 link-local address as a gateway (VIP) on the active gateway IPv6 configuration.

If VRRP or active forwarding is configured on an SVI, active gateway cannot be configured. Active gateway with overlapping networks is not allowed. Maximum of 16 unique virtual MACs are supported in a system.

The maximum number of supported active gateways per switch is 4,000. Since a maximum of 31 secondary IPv4 addresses can be configured on an SVI, 32 IPv4 active gateways (along with the primary IPv4 address) can be configured per SVI with IP multinetting support. This support is also the same for IPv6 addresses.



NOTE: Do not use peer system MAC address as an active-gateway VMAC. If same MAC address is used, the VSX synchronization will try to sync the configuration on secondary switch and cause traffic disruptions.

Examples

Configuring active-gateway when the IP address is different from the SVI IP address on both VSX peers (valid for IPv6 and IPv4):

Switch 1:

```
Switch1(config)# int vlan 10
Switch1(config-if-vlan)# vrf attach VRF1
Switch1(config-if-vlan)# ip address 10.1.10.2/24
Switch1(config-if-vlan)# active-gateway ip mac 12:00:00:00:01:00
Switch1(config-if-vlan)# active-gateway ip 10.1.10.1
Switch1(config-if-vlan)# ipv6 address fd00:10:1:10::2/64
Switch1(config-if-vlan)# active-gateway ipv6 mac 12:00:00:00:01:00
Switch1(config-if-vlan)# active-gateway ipv6 fe80::1
```

Switch 2:

```
Switch2(config)# interface vlan10
Switch2(config-if-vlan)# vrf attach VRF1
Switch2(config-if-vlan)# ip address 10.1.10.3/24
Switch2(config-if-vlan)# active-gateway ip mac 12:00:00:00:01:00
Switch2(config-if-vlan)# active-gateway ip 10.1.10.1
Switch2(config-if-vlan)# ipv6 address fd00:10:1:10::3/64
Switch2(config-if-vlan)# active-gateway ipv6 mac 12:00:00:00:01:00
Switch2(config-if-vlan)# active-gateway ipv6 fe80::1
```


Configuring active-gateway when the IP address is the same as the SVI IP address on both VSX peers (valid for IPv4 only):

Switch 1:

```
Switch1(config)# int vlan 10
Switch1(config-if-vlan)# vrf attach VRF1
Switch1(config-if-vlan)# ip address 10.1.10.2/24
Switch1(config-if-vlan)# active-gateway ip mac 12:00:00:00:01:00
Switch1(config-if-vlan)# active-gateway ip 10.1.10.1
```

Switch 2:

```
Switch2(config)# int vlan 10
Switch2(config-if-vlan)# vrf attach VRF1
Switch2(config-if-vlan)# ip address 10.1.10.3/24
Switch2(config-if-vlan)# active-gateway ip mac 12:00:00:00:01:00
Switch2(config-if-vlan)# active-gateway ip 10.1.10.1
```

Configuring only the active gateway address:

```
switch(config-if-vlan)# ip address 192.168.1.250/24
switch(config-if-vlan)# active-gateway ip 192.168.1.251
```

Configuring only the active gateway IP MAC address:

```
switch2(config-if-vlan)# ip address 192.168.1.250/24
switch2(config-if-vlan)# active-gateway ip mac 00:00:00:01:00:01
```

Removing the active gateway for active-active routing (IPv6 and IPv4):

```
switch(config-if-vlan)# no active-gateway ip
switch(config-if-vlan)# no active-gateway ipv6
```

Removing the active gateway for active-active routing for an IP address:

```
switch(config-if-vlan)# no active-gateway ip 192.168.1.250
```

Removing the active gateway for active-active routing for virtual MAC addresses:

```
switch(config-if-vlan)# no active-gateway ip mac
```

arp-suppression

Not supported on the 8400 Switch Series.

Syntax

```
arp-suppression
```

```
no arp-suppression
```

Description

Enables ARP suppression for EVPN-VXLAN globally across all Layer 2 VNIs configured on the VTEP. If the target address is present in the neighbor cache, the switch responds to the broadcast or unicast ARP request. ARP suppression is disabled by default. If the target IP/MAC is not present, the switch forwards arp request over the VXLAN data plane for neighbor resolution.

The `no` form of this command disables the ARP suppression.

Command context

config-evpn

Command context

config-evpn-vlan

Examples

Configuring ARP suppression in EVPN:

```
switch(config-evpn) # arp-suppression
switch(config-evpn) # no arp-suppression
```

evpn

Syntax

evpn

no evpn

Description

Specifies the EVPN context which provides the configurations for VLAN-based EVPN service mode. The **no** form of this command removes this configuration.

Command context

config-evpn

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the EVPN context:

```
switch(config) # evpn
switch(config-evpn) #
```

Removing the EVPN configuration context:

```
switch(config-evpn) # no evpn
```

mac-move-detection count timer

Syntax

mac-move-detection count <MAC-MOVE-COUNT> timer <MAC-MOVE-TIMER>

no mac-move-detection count <MAC-MOVE-COUNT> timer <MAC-MOVE-TIMER>

Description

Configures EVPN MAC dampening for duplicate MAC and MAC-move count and timer across VTEPs.

The **no** form of this command resets the value of the count and timer to the default values of 5 and 180 seconds respectively.



NOTE: EVPN MAC dampening is always enabled. Links to the VTEPs must be always up for EVPN MAC dampening to be activated.

Command context

config-evpn

Parameters

count <MAC-MOVE-COUNT>

Specifies the number of MAC-moves for MAC dampening to take effect. Range: 2 to 10. Default: 5.

timer <MAC-MOVE-TIMER>

Specifies the MAC-move time limit in seconds for MAC dampening to take effect. Range: 1 to 1000 seconds. Default: 180 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring EVPN MAC dampening:

```
switch(config-evpn) # mac-move-detection count 6 timer 199
```

The above command dampens a MAC if the MAC moves six times within 199 seconds.

```
switch(config-evpn) # mac-move-detection count 8
```

The above command dampens a MAC if the MAC moves eight times within 180 seconds.

```
switch(config-evpn) # mac-move-detection timer 255
```

The above command dampens a MAC if the MAC moves five times within 255 seconds.

nd-suppression

Not supported on the 8400 Switch Series.

Syntax

nd-suppression

no nd-suppression

Description

Enables ND suppression for EVPN-VXLAN globally. If the target address is present in the NDMD cache, the switch responds to the IPv6 multicast or unicast neighbor solicitation. ND suppression is disabled by default. This feature is not supported on 8400 platform.

The **no** form of this command disables the ND suppression.

Command context

config-evpn

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring ND suppression in EVPN:

```
switch(config-evpn)# nd-suppression  
switch(config-evpn)# no nd-suppression
```

rd

Syntax

```
rd {auto | <AS-NUMBER:ID> | <IP-ADDRESS:ID>}  
  
no rd {auto | <AS-NUMBER:ID> | <IP-ADDRESS:ID>}
```

Description

Specifies a unique number prepended to the advertised routes within the VLAN. It ensures support for overlapping IP addresses and MACs across different tenants. The default value is NULL. Route Distinguisher (RD) has to be manually configured by a user.

The **no** form of this command removes the currently configured value.

Command context

config-evpn-vlan

Parameters

auto

Specifies automatic route filtering.

<AS-NUMBER:ID>

Specifies the AS number. It can be a 1-byte or 4-byte value. If the AS number is a 2-byte value, the administrative number is a 4-byte value and if the AS number is 4-byte value, the administrative number is a 2-byte value.

<IP-ADDRESS:ID>

Specifies the IP address. It is a 4-byte value and the id is 2 bytes.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring Route Distinguisher for EVPN VLAN:

```
switch(config-evpn)# vlan 10  
switch(config-evpn-vlan-10)# rd 6800:1  
switch(config-evpn)# vlan 20  
switch(config-evpn-vlan-20)# rd auto
```

redistribute host-route

Syntax

```
redistribute host-route  
  
no redistribute host-route
```

Description

Enables type-2 route advertisement to include the L3VNI, RT, and router MAC of the associated IP-VRF. It is applicable only in Symmetric routing where L3VNI is configured.

The `no` form of this command disables the redistribution of host routes.

Command context

`config-evpn-vlan`

Parameters

`host-route`

Specifies redistribution of host routes.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring Redistribute host-route in EVPN:

```
switch(config-evpn)# vlan 10  
switch(config-evpn-vlan-10)# redistribute host-route
```

`redistribute local-svi`

Syntax

`redistribute local-svi`

`no redistribute local-svi`

Description

Enables type-2 route advertisement for the local IP address and MAC address of the SVI interfaces corresponding to the EVPN-enabled VLANs.

The `no` form of this command disables type-2 route advertisement for the local IP address and MAC address of the SVI interfaces corresponding to the EVPN-enabled VLANs.

Command context

`config-evpn`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling type-2 route advertisement:

```
switch(config)# evpn  
switch(config-evpn)# redistribute local-svi
```

Disabling type-2 route advertisement:

```
switch(config)# evpn  
switch(config-evpn)# no redistribute local-svi
```

route-target

Syntax

```
route-target {import | export | both} {auto | <AS-NUMBER:ID> | <IP-ADDRESS:ID>}  
  
no route-target {import | export | both} {auto | <AS-NUMBER:ID> | <IP-ADDRESS:ID>}
```

Description

Controls the import and export of VPN routes only to the systems in the network for which routes are needed. The default value is NULL. Route Targets (RT) have to be manually configured by a user.

The `no` form of this command removes the currently configured value.

Command context

config-evpn-vlan

Parameters

import

Configures the route-target to import EVPN routes.

export

Configures the route-target to export EVPN routes.

both

Configures the route-target to import and export EVPN routes.

auto

Specifies automatic route filtering.

<AS-NUMBER:ID>

Specifies the AS number. It can be a 1-byte or 4-byte value. If the AS number is a 2-byte value, the administrative number is a 4-byte value and if the AS number is 4-byte value, the administrative number is a 2-byte value.

<IP-ADDRESS:ID>

Specifies the IP address. It is a 4-byte value and the id is 2 bytes.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring Route Targets for EVPN VLAN:

```
switch(config-evpn)# vlan 10  
switch(config-evpn-vlan-10)# route-target import 6800:1  
switch(config-evpn-vlan-10)# route-target export 6800:1  
switch(config-evpn)# vlan 20  
switch(config-evpn-vlan-20)# route-target both 6900:1
```

route-target {evpn}

Syntax

```
route-target {import | export | both} {auto | <AS-NUMBER:ID> | <IP-ADDRESS:ID>} evpn  
no route-target {import | export | both} {auto | <AS-NUMBER:ID> | <IP-ADDRESS:ID>} evpn
```

Description

Configures the route target (RT) for EVPN VRF to control the import and export of VPN routes only to the systems in the network for which routes are needed. The default value is NULL. Route targets have to be manually configured by a user.

The `no` form of this command removes the RT in EVPN VRF.

Command context

```
config-vrf
```

Parameters

import

Imports the VRF routes that match the RT.

export

Exports the RT in the VRF routes.

both

Configures both import and export of routes for the VRF.

auto

Specifies automatic route filtering.

<AS-NUMBER: ID>

Specifies the AS number. It can be a 1-byte or 4-byte value. If the AS number is a 2-byte value, the administrative number is a 4-byte value. If the AS number is 4-byte value, the administrative number is a 2-byte value.

<IP-ADDRESS: ID>

Specifies the IP address. It is a 4-byte value and the id is 2 byte value.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring Route Targets for EVPN VRF:

```
switch(config-vrf)# route-target import 6800:1 evpn  
switch(config-vrf)# route-target export 6800:1 evpn  
switch(config-vrf)# route-target both 6800:1 evpn
```

show evpn evi

Syntax

```
show evpn evi
```

Description

Shows the information of EVPN instances.

Command context

Operator (>) or Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing information for EVPN instances:

```
switch# show evpn evi

L2VNI : 100
  Route Distinguisher : 10.10.10.1:10
  VLAN                : 10
  Status              : Up
  RT Import           : 1.1.1.1:1, 2.2.2.2:1, 3.3.3.3:1, 5:1
  RT Export           : 4.4.4.4:61, 1000:21
  Local MACs          : 30
  Remote MACs         : 945
  Peer VTEPs          : 8

L2VNI : 200
  Route Distinguisher :
  VLAN                : 20
  Status              : Down, No RD
  RT Import           : 1.1.1.1:2, 2.2.2.2:2, 3.3.3.3:2, 4.4.4.4:2,
                      5.5.5.5:2, 5:2
  RT Export           : 4.4.4.4:62, 1000:22
  Local MACs          :
  Remote MACs         :
  Peer VTEPs          :

L2VNI : 300
  Route Distinguisher : 10.10.10.1:30
  VLAN                : 30
  Status              : Up
  RT Import           : 1.1.1.1:3, 2.2.2.2:3, 3.3.3.3:3, 5:3
  RT Export           : 4.4.4.4:63, 1000:23
  Local MACs          : 30
  Remote MACs         : 945
  Peer VTEPs          : 12

L3VNI : 1000
  Route Distinguisher : 10.10.10.1:1000
  VRF                 : vrf1000
  Status              : Up
  RT Import           : 1.1.1.1:4, 2.2.2.2:4, 3.3.3.3:4, 5:4
  RT Export           : 4.4.4.4:64, 1000:24
  Local Type-5 Routes : 2
  Remote Type-5 Routes : 3
  Peer VTEPs          : 6
```

show evpn evi summary

Syntax

show evpn evi summary

Description

Shows the summary information for EVPN instances.

Command context

Operator (>) or Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing summary information for EVPN instances:

```
switch# show evpn evi summary
```

L2VNI	VLAN	Status
100	10	Up
200	20	Down, RT conflict
210	21	Up
220	22	Down, No RT
230	23	Down, No RT
240	24	Up
250	25	Up
260	26	Up
270	27	Up
280	28	Up
290	29	Up
310	31	Up
L3VNI	VRF	Status
1000	vrf1000	Up
1001	vrf1001	Down, RT conflict
1002	vrf1002	Down, No RD
1004	vrf1003	Down, Administratively down
1005	vrf1003	Up
EVPN instances		: 17
EVPN instances Up		: 11

```
show evpn evi <EVI-ID>
```

Syntax

```
show evpn evi <EVI-ID>
```

Description

Shows the information for the particular EVPN instance.

Command context

Operator (>) or Manager (#)

Parameter

<EVI-ID>

Specifies the EVPN instance ID.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing information for the particular EVPN instance:

```
switch# show evpn evi 100
L2VNI : 100
  Route Distinguisher      : 10.10.10.1:10
  VLAN                    : 10
  Status                  : Up
  RT Import               : 1.1.1.1:1, 2.2.2.2:1, 3.3.3.3:1, 5:1
  RT Export               : 4.4.4.4:61, 1000:21
  Local MACs              : 30
  Remote MACs             : 945
  Peer VTEPs              : 8
```

show evpn evi detail

Syntax

```
show evpn evi detail
```

Description

Shows the detailed information for all EVPN instances.

Command context

Operator (>) or Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing detailed information for EVPN instances:

```
switch# show evpn evi detail
L2VNI : 100
  Route Distinguisher      : 10.10.10.1:10
  VLAN                    : 10
  Status                  : Up
  RT Import               : 1.1.1.1:1, 2.2.2.2:1, 3.3.3.3:1, 5:1
  RT Export               : 4.4.4.4:61, 1000:21
  Local MACs              : 30
  Remote MACs             : 307
  Peer VTEPs              : 8

  Peer VTEPs              Remote MACs
  -----
  10.10.10.2              40
  10.10.10.3              22
  10.10.10.4              15
  10.10.10.5              155
  10.10.10.6              25
  10.10.10.7              35
  10.10.10.8              50
  10.10.10.9              55
```

```

L2VNI : 200
Route Distinguisher      :
VLAN                     : 20
Status                   : Down, No RD
RT Import                 : 1.1.1.1:2, 2.2.2.2:2, 3.3.3.3:2, 4.4.4.4:2,
                          5.5.5.5:2, 5:2
RT Export                 : 4.4.4.4:62, 1000:22
Local MACs                :
Remote MACs               :
Peer VTEPs                :

```

Peer VTEPs	Remote MACs

```

L2VNI : 300
Route Distinguisher      : 10.10.10.1:30
VLAN                     : 30
Status                   : Up
RT Import                 : 1.1.1.1:3, 2.2.2.2:3, 3.3.3.3:3, 5:3
RT Export                 : 4.4.4.4:63, 1000:23
Local MACs                : 30
Remote MACs               : 362
Peer VTEPs                : 12

```

Peer VTEPs	Remote MACs

10.10.10.2	60
10.10.10.3	12
10.10.10.4	13
10.10.10.5	15
10.10.10.6	15
10.10.10.7	35
10.10.10.8	53
10.10.10.9	45
10.10.10.10	11
10.10.10.11	12
10.10.10.12	35
10.10.10.13	56

```

L3VNI : 1000
Route Distinguisher      : 10.10.10.1:1000
VRF                       : vrf1000
Status                   : Up
RT Import                 : 1.1.1.1:4, 2.2.2.2:4, 3.3.3.3:4, 5:4
RT Export                 : 4.4.4.4:64, 1000:24
Local Type-5 Routes      : 2
Remote Type-5 Routes     : 30
Peer VTEPs                : 12

```

Peer VTEPs	Remote Type-5 Routes

10.10.10.2	2
10.10.10.3	1
10.10.10.4	1
10.10.10.5	1
10.10.10.6	1
10.10.10.7	3
10.10.10.8	5
10.10.10.9	4
10.10.10.10	1
10.10.10.11	3
10.10.10.12	3
10.10.10.13	5

show evpn evi <EVI-ID> detail

Syntax

```
show evpn evi <EVI-ID> detail
```

Description

Shows the detailed information for the particular EVPN instance.

Command context

Operator (>) or Manager (#)

Parameter

<EVI-ID>

Specifies the EVPN instance ID.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing detailed information for the particular EVPN instance:

```
switch# show evpn evi 100 detail
L2VNI : 100
  Route Distinguisher      : 10.10.10.1:10
  VLAN                    : 10
  Status                   : Up
  RT Import                : 1.1.1.1:1, 2.2.2.2:1, 3.3.3.3:1, 5:1
  RT Export                : 4.4.4.4:61, 1000:21
  Local MACs               : 30
  Remote MACs              : 397
  Peer VTEPs               : 8

  Peer VTEPs               Remote MACs
  -----
  10.10.10.2                40
  10.10.10.3                22
  10.10.10.4                15
  10.10.10.5                155
  10.10.10.6                25
  10.10.10.7                35
  10.10.10.8                50
  10.10.10.9                55
```

show evpn mac-ip

Syntax

```
show evpn mac-ip
```

Description

Show the information about the EVPN MAC-IP for the EVPN instances.

Command context

Operator (>) or Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing information about the EVPN MAC-IP for the EVPN instances:

```
switch# show evpn mac-ip
Flags: Local(L), Remote(R)
```

EVI	MAC	IP	Next-hop	Seq-Num	Flags
100	14:50:56:96:76:56		vxlan1(11.1.1.3)	0	R
100	14:50:56:96:76:56	3.3.4.5	vxlan1(11.1.1.3)	0	R
100	14:50:56:96:76:56	3.3.5.5	vxlan1(11.1.1.3)	0	R
100	24:50:56:96:76:56	3.3.3.2	vxlan1(11.1.1.3)	1	R
100	34:50:56:96:76:56	3.3.6.2		2	L
100	44:50:56:96:76:56	3.3.7.3		2	L
200	52:50:56:96:76:56	5.5.5.2		0	L

```
MACs : 5
Remote MACs : 2
```

show evpn mac-ip evi

Syntax

```
show evpn mac-ip evi <EVI-ID>
```

Description

Show the information about the EVPN MAC-IP for the particular EVPN instance.

Command context

Operator (>) or Manager (#)

Parameter

<EVI-ID>

Specifies the EVPN instance ID.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing information about the EVPN MAC-IP for the particular EVPN instance:

```
switch# show evpn mac-ip evi 100
Flags: Local(L), Remote(R)
```

MAC	IP	Next-hop	Seq-Num	Flags
14:50:56:96:76:56		vxlan1(11.1.1.2)	0	R
14:50:56:96:76:56	3.3.4.5	vxlan1(11.1.1.3)	0	R
14:50:56:96:76:56	3.3.5.4	vxlan1(11.1.1.2)	0	R
24:50:56:96:76:56	3.3.3.2	vxlan1(11.1.1.3)	1	R
34:50:56:96:76:56	3.3.6.2		2	L
44:50:56:96:76:56	3.3.7.3		2	L

MACs : 4
Remote MACs : 2

show evpn vtep-neighbor

Syntax

```
show evpn vtep-neighbor {all-vrfs | vrf <VRF-Name>}
```

Description

Shows the remote VTEPs MAC-IP binding. The state of the peer VTEP denotes whether VXLAN tunnel to the VTEP is Up or Down.

Command context

Operator (>) or Manager (#)

Parameters

all-vrfs

Display information for all VRFs.

vrf <vrf-name>

Specify a VRF by VRF name (if no <VRF-NAME> is specified, the default VRF is implied).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing EVPN VTEP neighbor information for all VRFs:

```
switch# show evpn vtep-neighbor all-vrfs
VTEP-IP      L3VNI  MAC                VRF      State
-----
2.2.2.2      1234   00:20:56:bd:27:bc  VRF1234  Up
2.2.2.2      6789   00:20:56:bd:27:bc  VRF6789  Up
3.3.3.3      1234   00:30:56:ef:aa:cc  VRF1234  Down
4.4.4.4      6789   00:40:56:12:34:44  VRF6789  Up
5.5.5.5      6789   00:50:56:ab:11:ee  VRF6789  Up
```

Showing EVPN VTEP neighbor information for the specified VRF name:

```
switch# show evpn vtep-neighbor vrf VRF1234
VTEP-IP      L3VNI  MAC                VRF      State
-----
2.2.2.2      1234   00:20:56:bd:27:bc  VRF1234  Up
3.3.3.3      1234   00:30:56:ef:aa:cc  VRF1234  Down
```

show running-config evpn

Syntax

```
show running-config evpn
```

Description

Shows all EVPN configurations.

Command context

Operator (>) or Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing all EVPN configurations:

```
switch# show running-config evpn
evpn
  vlan 10
    rd 6800:1
    route-target import 6800:1
    route-target export 6800:1
  vlan 20
    rd 6900:1
    route-target import 6900:1
    route-target export 6900:1
```

virtual-mac

Syntax

```
virtual-mac <MAC-ADDR>
```

```
no virtual-mac
```

Description

Configures the virtual MAC address for EVPN.

The `no` form of this command removes the virtual MAC address configuration.

Command context

```
config
```

Parameters

<MAC-ADDR>

Specifies the virtual MAC address.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring virtual MAC address for EVPN:

```
switch(config)# virtual-mac ab:12:33:33:03:22
switch(config)# no virtual-mac
```



NOTE: For EVPN symmetric IRB to work, `virtual-mac` must be configured and it must be unique for all the VTEPs involved in EVPN except for VSX nodes. In the VSX pair switches, same `virtual-mac` must be configured in both the VTEPs. In case of VSX VTEP, it is recommended to use the same value as the VSX `system-mac` value in order to simplify the troubleshooting.

vlan

Syntax

```
vlan <ID>
```

```
no vlan <ID>
```

Description

Specifies the VLAN ID and enters the VLAN context under EVPN.

The `no` form of this command removes this configuration.

Command context

```
config-evpn-vlan
```

Parameters

<ID>

Specifies the VLAN ID. Range: 2 - 4040.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config-evpn)# vlan 10  
switch(config-evpn-vlan-10)#
```


interface vxlan

Syntax

```
interface vxlan 1
```

```
no interface vxlan 1
```

Description

Creates VXLAN interface 1 and changes to the `config-vxlan-if` context. A maximum of one VXLAN interface is supported. By default, the VXLAN is disabled. To enable the VXLAN, use the command `no shutdown`.

The `no` form of this command removes VXLAN interface 1. This deletes the VXLAN tunnel, and all VNIs and VLAN-to-VNI mappings associated with it.

Command context

```
config
```

Parameters

1

Only one VXLAN interface is supported.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating VXLAN interface 1:

```
switch(config)# interface vxlan 1  
switch(config-vxlan-if)#
```

Deleting VXLAN interface 1:

```
switch(config)# no interface vxlan 1
```

routing

Syntax

```
vni <VNI-NUMBER>
```

```
no vni <VNI-NUMBER>
```

Description

Enables a layer 3 VNI for EVPN VXLAN distributed L3 gateways with symmetric IRB. The VNI is automatically assigned the default VRF. To assign another VRF, use the command `vrf`. Used with EVPN-based VXLANs.

The `no` form of this command disables symmetric routing on a VNI.

Command context

```
config-vni-<VNI-NUMBER>
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling L3 VNI 1000 for EVPN VXLAN distributed L3 gateways with symmetric IRB using VRF `vrf-1`:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# routing
switch(config-vni-1000)# vrf vrf-1
```

Disabling L3 VNI 1000 for EVPN VXLAN distributed L3 gateways with symmetric IRB:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# no routing
```

show interface vxlan1

Syntax

```
show interface vxlan1
```

Description

Shows detailed information about VXLAN interface 1.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

On the 6400 Switch Series, interface identification differs.

Showing settings for VXLAN interface 1.

```
switch# show interface vxlan1
Interface vxlan1 is up
Admin state is up
Description:VXLAN1
Underlay VRF: default
Destination UDP port: 4789
VTEP source IPv4 address: 1.1.1.1
```

VNI	Routing	VLAN	VRF	VTEP Peers	Origin
10	disabled	10	--	2.2.2.2	static
10	disabled	10	--	3.3.3.3	static
10	disabled	10	--	4.4.4.4	static
20	disabled	20	--	5.5.5.5	evpn
20	disabled	20	--	6.6.6.6	evpn
30	disabled	--	--	--	static
40	disabled	40	--	--	static
50	disabled	--	--	7.7.7.7	static
4000	enabled	--	default	22.1.1.2	evpn
4001	enabled	--	default	23.1.1.3	evpn

Aggregate Statistics

```
-----
Decap:
    104222 input packets          15841744 bytes
      236 broadcast packets      26942 bytes
        0 drop packets

Encap:
    108527 output packets        11068728 bytes
        6 BUM packets           422 bytes
        0 drop packets
```

show interface vxlan vni

Syntax

```
show interface vxlan vni <VNI-NUMBER> vteps
```

Description

Shows detailed VNI information for VXLAN interface 1.

Command context

Manager (#)

Parameters

<VNI-NUMBER>

Specifies the number for a VNI. Range: 1 to 16777214.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Status can be one of the following:

- operational: Virtual network ID is fully programmed on the switch hardware.
- configuration_error: Virtual network ID programming in the switch hardware failed due to misconfiguration.
- no_hw_resources: Virtual network ID programming failed in the switch hardware due to insufficient resources.
- activating: Initial state of virtual network ID when it is configured.

When a tunnel endpoint is directly connected via nexthop, then nexthop reachability appears empty (--).

Example

Showing VNI information:

```
switch# show interface vxlan vni
VNI      : 1000          VLAN   : 10
Routing  : disabled     VRF    : --
VNI-Status : operational

VNI      : 2000          VLAN   : 20
Routing  : enabled      VRF    : default
VNI-Status : activating
...
...

switch# show interface vxlan vni 1000
VNI      : 1000          VLAN   : 10
Routing  : disabled     VRF    : --
VNI-Status : operational
```

```
switch# show interface vxlan vni vteps
VNI      : 1000          VLAN   : 10
Routing  : disabled     VRF    : --
VNI-Status : operational
VTEPS
=====
  ORIGIN      SOURCE      DESTINATION  VRF      VTEP-STATUS
  -
  static      11.0.0.1      11.0.0.2    default  operational

VNI      : 2000          VLAN   : 20
Routing  : enabled      VRF    : default
VNI-Status : operational
VTEPS
=====
  ORIGIN      SOURCE      DESTINATION  VRF      VTEP-STATUS
  -
  evpn        11.0.0.1      14.0.0.2    default  activating
```

show interface vxlan vteps

Syntax

```
show interface vxlan vteps [detail | <IPV4-ADDR>]
```

Description

Shows information about the VTEPs on VXLAN interface 1.

Command context

Manager (#)

Parameters

detail

Show detailed information.

<IPV4-ADDR>

Specifies the IP address of a VTEP peer in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Status can be one of the following:

- operational: Tunnel endpoint is fully programmed on the switch hardware.
- configuration_error: Tunnel endpoint programming in the switch hardware failed due to misconfiguration.
- no_hw_resources: Tunnel endpoint programming failed in the switch hardware due to insufficient resources.
- activating: Initial state of tunnel endpoint when it is configured.

When a tunnel endpoint is directly connected via nexthop, then nexthop reachability appears empty (--).

Example

On the 6400 Switch Series, interface identification differs.

Showing information for the VTEPs on VXLAN interface 1.

switch# show interface vxlan vteps							
Source	Destination	Origin	Status	VNI	Routing	VLAN	VRF
11.0.0.1	11.0.0.2	static	operational	1000	disabled	10	--
11.0.0.1	12.0.0.2	static	activating	2000	disabled	20	--
11.0.0.1	22.1.1.1	evpn	operational	4000	enabled	--	red
11.0.0.1	23.1.1.1	evpn	activating	4001	enabled	--	blue

Showing detailed information for the VTEPs on VXLAN interface 1.

```
switch# show interface vxlan vteps detail
Destination : 22.22.22.1
Source      : 21.21.21.1
Origin      : static
VRF         : default
Status      : operational
Nexthops
=====
  IP-ADDRESS  INTERFACE  NEXTHOP-MAC
  -----
    --        1/1/2        11:11:11:11:33:11

Destination : 33.33.33.1
Source      : 21.21.21.1
Origin      : evpn
VRF         : default
Status      : operational
Nexthops
=====
  IP-ADDRESS  INTERFACE  NEXTHOP-MAC
  -----
```

```

2.2.3.1      1/1/1      11:11:11:11:44:11
2.2.2.1      lag1      11:11:11:11:22:11
2.2.1.1      vlan21    11:11:11:11:11:11
...
...
switch# show interface vxlan vteps 33.33.33.1
Destination   : 33.33.33.1
Source        : 21.21.21.1
Origin        : evpn
VRF           : default
Status        : operational
Nexthops
=====
      IP-ADDRESS  INTERFACE  NEXTHOP-MAC
-----
2.2.3.1      1/1/1      11:11:11:11:44:11
2.2.2.1      lag1      11:11:11:11:22:11
2.2.1.1      vlan21    11:11:11:11:11:11

```

shutdown

Syntax

```
shutdown
```

```
no shutdown
```

Description

Disables the VXLAN interface and deletes all VXLAN tunnels, segments, and members on the interface.

The `no` form of this command starts the VXLAN interface and creates all VXLAN tunnels and segments. If members are configured, they are added to the VXLAN segment.

Command context

```
config-vxlan-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling VXLAN interface 1:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# shutdown
```

Enabling VXLAN interface 1:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# no shutdown
```

source ip

Syntax

```
source ip <IPV4-ADDR>
```

Description

Configures the source IP address for a VXLAN interface. All VXLAN encapsulated packets use this source IP address in the outer IP header.

If you change an existing source IP address, all tunnels with the old source IP address are deleted, and new tunnels are created with the new source IP address.

The `no` form of this command deletes the source IP address for the VXLAN interface and deletes all VXLAN tunnels using this source IP address.

Command context

```
config-vxlan-if
```

Parameters

<IPV4-ADDR>

Specifies the IP address to assign to the VXLAN interface in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. This must be an address assigned to an existing switch interface, either a loopback interface or a layer 3 interface.

Authority

Administrators or local user group members with execution rights for this command.

Examples

On the 6400 Switch Series, interface identification differs.

Configuring the loopback IP address as the source IP address:

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 1.1.1.1/24
switch(config)# interface vxlan 1
switch(config-vxlan-if)# source ip 1.1.1.1
```

Configuring a layer 3 interface IP address as the source IP address:

```
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# routing (6300/6400 only)
switch(config-if)# ip address 11.10.10.1/24
switch(config)# interface vxlan 1
switch(config-vxlan-if)# source ip 10.10.10.1
```

Deleting the source IP address for VXLAN interface 1:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# no source ip 10.10.10.1
```

system vlan-client-presence-detect

Supported on the 6300 and 6400 Switch Series only.

Syntax

```
system vlan-client-presence-detect
```

```
no system vlan-client-presence-detect
```

Description

Enables VNI mapped VLANs when detecting the presence of a client. When enabled, VNI mapped VLANs are *up* only if there are authenticated clients on the VLAN, or if the VLAN has statically configured ports and those ports are *up*. When not enabled, VNI mapped VLANs are always *up*.

The `no` form of this command disables detection of clients on VNI mapped VLANs.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling detection of clients:

```
switch(config)# system vlan-client-presence-detect
```

Disabling detection of clients:

```
switch(config)# no system vlan-client-presence-detect
```

vlan

Syntax

```
vlan <VLAN-ID>
```

```
no vlan <VLAN-ID>
```

Description

Associates an existing VLAN with a VNI. Only one VLAN can be associated with a VNI and the VNI must have symmetric routing disabled. To change the VLAN associated with a VNI, execute the command `vlan` with a different VLAN ID.

The `no` form of this command removes the specified VLAN from a VNI. Traffic on the specified VLAN is no longer bridged on the VXLAN interface.

Command context

```
config-vni-<VNI-NUMBER>
```

Parameters

<VLAN-ID>

Specifies the number of an existing VLAN. Range: 2 to 4040.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning VLAN 10 to VNI 1000:


```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# vlan 10
```

Deleting VLAN 10 from VNI 1000:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# no vlan 10
```

vni

Syntax

```
vni <VNI-NUMBER>
```

```
no vni <VNI-NUMBER>
```

Description

Creates a VNI (Virtual Network Identifier) for the VXLAN interface and changes to the `config-vni-<VNI-NUMBER>` context. The VNI identifies a VXLAN segment, which acts as a logical network. The VNI can be associated with either a VLAN or VRF.

- When the VNI is associated with a VLAN, the VNI supports asymmetric routing.
- Enable support for asymmetric routing by executing the `routing` command. By default, the VNI is associated with the default VRF. To use another VRF, execute the `vrf` command.

The `no` form of this command deletes the specified VNI from the VXLAN interface. All VXLAN tunnels, VXLAN segments, and members associated with the VNI are deleted.

Command context

```
config-vxlan-if
```

Parameters

<VNI-NUMBER>

Specifies the number for a VNI. Range: 1 to 16777214.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating VNI 1000:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)#
```

Deleting VNI 1000:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# no vni 1000
```

vrf

Syntax

```
vrf <VRF-NAME>
```

```
no vrf <VRF-NAME>
```

Description

Changes the VRF associated with an L3 VNI after symmetric routing is activated using the `routing` command. The default VRF should not be configured as an EVPN-enabled VRF.

The `no` form of this command sets the VRF associated with an L3 VNI to the default VRF.

Command context

```
config-vni-<VNI-NUMBER>
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling L3 VNI 1000 using VRF `vrf-1`:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# routing
switch(config-vni-1000)# vrf vrf-1
```

Setting the VRF on L3 VNI 1000 to the default VRF:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# routing
switch(config-vni-1000)# no vrf vrf-1
```

vtep-peer

Syntax

```
vtep-peer <IPV4-ADDR>
```

```
no vtep-peer <IPV4-ADDR>
```

Description

Adds a VTEP peer to a VNI. The VNI must not have routing enabled. The VTEP peer IP address must be reachable for a VXLAN tunnel to be established.

The `no` form of this command removes a VTEP peer from a VNI, which deletes the VXLAN tunnel to the peer.

Command context

```
config-vni-<VNI-NUMBER>
```

Parameters

<IPV4-ADDR>

Specifies the IP address of a VTEP peer in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding VTEP peer 10.10.10.1 to VNI 1000:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# vlan 10
switch(config-vni-1000)# vtep-peer 10.10.10.1
```

Deleting VTEP peer 10.10.10.1 from VNI 1000:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# no vtep-peer 10.10.10.1
```

vxlan-counters aggregate

Supported on the 8325 and 8360 Switch Series only. Enabled by default in the 8400 Switch Series.

Syntax

```
vxlan-counters aggregate
```

```
no vxlan-counters aggregate
```

Description

Attaches VXLAN counters to a VXLAN interface. The counters aggregate statistics for packets sent through the interface. Display statistics with the command `show interface vxlan`. Statistics are only displayed once a valid configuration is made on the interface.

Command context

```
config-vxlan-if
```

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Enabling counters for VXLAN interface 1.

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vxlan-counters aggregate
```

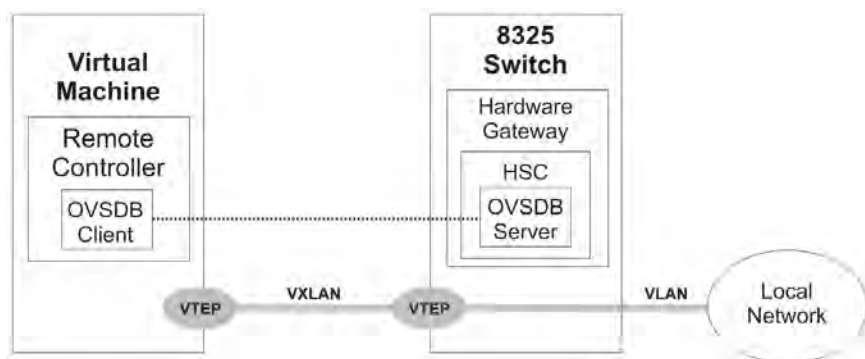
Disabling counters for VXLAN interface 1.

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# no vxlan-counters aggregate
```

Supported on the 8325, 8360, and 8400 Switch Series only.

Overview

The switch can function as a hardware layer 2 gateway, which enables the switch to extend VXLAN networks to a remote hypervisor VTEP. Both the switch and hypervisor VTEP are controlled by a centralized controller (such as VMware NSX-V 6.4.4).



The hardware layer 2 gateway encapsulates or decapsulates VXLAN traffic between the virtual machine and VLANs connected to the switch.

Communication between the remote controller and the hardware layer 2 gateway is managed by the remote controller. It communicates with a software module on the switch called the hardware switch controller (HSC). The HSC runs an open source OVSDB server, and the remote controller connects to it as an OVSDB client. OVSDB is the Open vSwitch Database Management Protocol detailed in RFC 7047.

Limitations

- HSC is mutually exclusive with EVPN.
- Broadcast, Unknown unicast and Multicast (BUM) traffic is not replicated by HSC. Instead, the NSX Controller provides a list of Service Replication Nodes (RSNs) that the HSC will leverage for replication.

Connecting to a remote controller

Procedure

1. Create an HSC with the command **`hsc`**.
2. Configure the IP address of the HSC manager on the remote controller with the command **`manager ip`**. The HSC manager must be reachable via the management port on the switch.
3. Optionally, change the TCP port on which the HSC communicates with the HSC manager with the command **`manager port`**. By default, port 6640 is used. The remote controller OVSDB client listens on this default port.

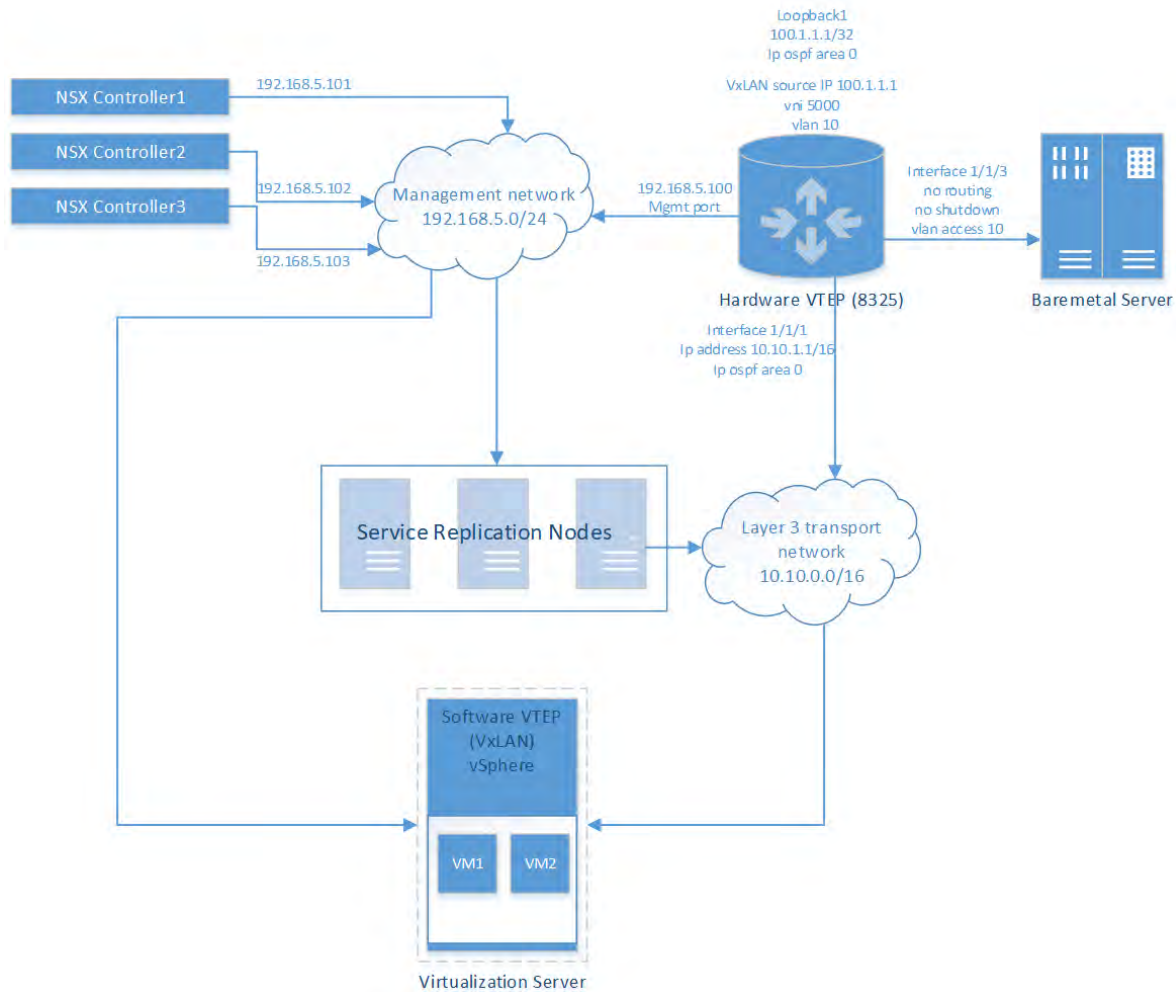
4. Enable the HSC with the command **enable**.
5. Obtain the default certificate on the HSC with the command `show crypto pki certificate local-cert pem`. Copy the lines between **-----BEGIN CERTIFICATE-----** and **-----END CERTIFICATE-----**. You need to provide this certificate when configuring the remote controller.

Example

```
switch(config)# hsc
switch(config-hsc)# enable
Trust Anchor (TA) profile configurations are not used for HSC
authentication. Instead HSC will store the CA certificate from the
HSC controller during the first TLS handshake and use it for all
future authentications with HSC servers. This CA certificate will
not be used for any other certificate-based authentication. Do you
want to continue (y/n)? y
switch(config-hsc)# exit
switch(config)# exit
switch# show crypto pki certificate local-cert pem
Certificate name: local-cert
Associated Applications:
  captive-portal, hsc, https-server, syslog-client
Certificate status: installed
Certificate type: self-signed
-----BEGIN CERTIFICATE-----
MIDITCDskKkeLkDKfjlsafkdjLdfkejwlisfuslekfjsdkfjelrjsekfslkefjsselfkjslde8383
...
3md0k4o9vjksdoijeknkviocvhsksdoeo399((jifiIIHFKwlielId8rekILF:IofJe,kei(gfo9
-----END CERTIFICATE-----
```

Scenario 1

This example illustrates how an 8325 switch can be connected to a VMware NSX.



Key components

- NSX controllers: Central control point for all logical switches in the network.
- Management network: Network on which the 8325 switch communicates with the NSX controllers.
- Hardware VTEP (8325 switch): Communication with the NSX controller occurs via the switch management port (which operates as a DHCP client by default). Interface 1/1/1 links the switch to the layer 3 network. Interface 1/1/3 extends the layer 2 domain over a VXLAN and links the switch to the bare metal server. OSPF is used to enable the routed layer 3 underlay network.
- Bare metal server: Physical server providing network services.
- Layer 3 transport network: The underlay network which provides routing functionality.
- Virtualization server: The virtualization server is managed by VMWare vSphere. It hosts software VTEPs which perform VXLAN encapsulation for VMs deployed in virtual servers (such as an ESXi server).

Procedure

1. On the 8325 switch:
 - a. Enable interface 1/1/1, assign the IP address 100.1.1.1/24 to it, and configure OSPF.

```
switch# config
switch(config)# interface 1/1/1
switch(config-if)# ip address 100.1.1.10/24
switch(config-if)# ip ospf area 0
switch(config-if)# no shutdown
switch(config-if)# exit
switch(config)# router ospf 1
switch(config-ospf)# router-id 1.1.1.1
switch(config-ospf)# router-id 1.1.1.1
switch(config-ospf)# router-id 1.1.1.1
```

- b. Create VLAN 10.

```
switch(config)# vlan 10
switch(config-vlan-10)# exit
```

- c. Create a loopback interface 1 with IP address 100.1.1.1/32 and assign it to OSPF area 0 on the default VRF.

```
switch(config)# router ospf 1 vrf default
switch(config-ospf-1)# exit
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 100.1.1.1/32
switch(config-loopback-if)# ip ospf 1 area 0
switch(config-loopback-if)# exit
switch(config)#
```

- d. Enable interface 1/1/3 and assign VLAN 10 to it.

```
switch(config)# interface 1/1/3
switch(config-if)# no shutdown
switch(config-if)# no routing
switch(config-if)# vlan access 10
switch(config-if)# exit
```

- e. Create VXLAN interface 1 and assign the source IP address 100.1.1.1/24 to it.

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# source ip 100.1.1.1
switch(config-vxlan-if)# no shutdown
```

- f. Create VNI 5000, and assign VLAN 10 to it.

```
switch(config-vxlan-if)# vni 5000
switch(config-vni)# vlan 10
switch(config-vni)# exit
switch(config-vxlan-if)# exit
```

- g. Configure the HSC manager address.

```
switch(config)# hsc
switch(config-hsc)# manager ip 10.10.10.1
```

- h. Enable the HSC. You are prompted to use the default certificate installed on the switch to secure the connection with the manager. Type **y**.

```
switch(config-hsc)# enable
Trust Anchor (TA) profile configurations are not used for HSC
authentication. Instead HSC will store the CA certificate from the
HSC controller during the first TLS handshake and use it for all
future authentications with HSC servers. This CA certificate will
```

not be used for any other certificate-based authentication. Do you want to continue (y/n)? **y**

- i. Make a copy of the default certificate (local-cert) in PEM format. You need to configure this certificate on the NSX manager.

```
switch# show crypto pki certificate local-cert pem
Certificate name: local-cert
Associated Applications:
  captive-portal, hsc, https-server, syslog-client
Certificate status: installed
Certificate type: self-signed
-----BEGIN CERTIFICATE-----
MIDITCDskKkeLkDKfjlsafkdjLdfkejwlisfuslekfjsdkfjelfrjsekfslkefjsselfkjslde8383
...
3md0k4o9vjksdoijeknkviochvshksdoeo399((jifiIIHFKwLIelId8rekILF:IofJe,kei(gfo9
-----END CERTIFICATE-----
```

Copy the lines between -----BEGIN CERTIFICATE----- and -----END CERTIFICATE-----.

2. On the NSX manager, add the switch certificate using the vSphere Web Client. Select **Networking & Security** and then **Service Definitions**. On the **Hardware Devices** tab, click **Add (+)** and paste the certificate into the appropriate box.

```
switch# show crypto pki certificate local-cert pem
Certificate name: local-cert
Associated Applications:
  captive-portal, hsc, https-server, syslog-client
Certificate status: installed
Certificate type: self-signed
-----BEGIN CERTIFICATE-----
MIDITCDskKkeLkDKfjlsafkdjLdfkejwlisfuslekfjsdkfjelfrjsekfslkefjsselfkjslde8383
...
3md0k4o9vjksdoijeknkviochvshksdoeo399((jifiIIHFKwLIelId8rekILF:IofJe,kei(gfo9
-----END CERTIFICATE-----
```

For more information on configuring the NSX manager, refer to the VMware NSX documentation.

HSC commands

bfd enable

Syntax

bfd enable

no bfd enable

Description

Enables BFD support on an HSC.

The `no` form of this command disables BFD support on an HSC.

Command context

config-hsc

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling BFD:

```
switch(config-hsc) # bfd enable
```

Disabling BFD:

```
switch(config-hsc) # no bfd enable
```

bfd disable

Syntax

```
bfd disable
```

Description

Disables BFD support on an HSC.

Command context

```
config-hsc
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling BFD:

```
switch(config-hsc) # bfd disable
```

disable

Syntax

```
disable
```

Description

Disables a hardware switch controller (HSC).

Command context

```
config-hsc
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling an HSC:

```
switch(config) # hsc  
switch(config-hsc) # disable
```

enable

Syntax

enable

no enable

Description

Enables a hardware switch controller (HSC).

The `no` form of this command disables an HSC.

Command context

config-hsc

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling an HSC:

```
switch(config)# hsc
switch(config-hsc)# enable
Trust Anchor (TA) profile configurations are not used for HSC
authentication. Instead HSC will store the CA certificate from the
HSC controller during the first TLS handshake and use it for all
future authentications with HSC servers. This CA certificate will
not be used for any other certificate-based authentication. Do you
want to continue (y/n)? y
```

Disabling an HSC:

```
switch(config)# hsc
switch(config-hsc)# no enable
```

hsc

Syntax

hsc

no hsc

Description

Creates a hardware switch controller (HSC) and changes to the `config-hsc` context for it. The switch uses the HSC to communicate with a remote controller (such as VMware NSX) to create an extended a layer 2 VXLAN network.

The HSC runs an open-source OVSDb server. OVSDb is the Open vSwitch Database Management Protocol detailed in RFC 7047. The remote controller connects to the server as an OVSDb client, and is responsible for managing the interaction with the HSC. As a result, the remote controller is known as the HSC manager.

The `no` form of this command removes the HDC controller and deletes all configuration settings.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating an HSC:

```
switch(config)# hsc
switch(config-hsc)#
```

Deleting an HSC:

```
switch(config)# no hsc
HSC configuration will be removed.
Do you want to continue (y/n)? y
```

manager ip

Syntax

manager ip <IPv4-ADDR>

no manager ip

Description

Configures the IP address of the HSC manager. The switch can connect with up to three managers. The **no** form of this command deletes the IP address of the HSC manager.



NOTE: HSC configuration can only be modified when the HSC is disabled.

Command context

config-hsc

Parameters

<IPv4-ADDR>

Specifies the IP address of the HSC manager in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. The IP address must be reachable on the management VRF.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the IP address of the HSC manager:

```
switch(config)# hsc
switch(config-hsc)# manager ip 1.1.1.1
```

Deleting the HSC manager IP address:

```
switch(config)# hsc
switch(config-hsc)# disable
switch(config-hsc)# no manager ip
```

manager port

Syntax

```
manager port <PORT-NUM>
```

```
no manager port
```

Description

Configures the TCP port on which the HSC communicates with the HSC manager.

The `no` form of this command sets the management port to the default value (6640).



NOTE: HSC configuration can only be modified when the HSC is disabled.

Command context

```
config-hsc
```

Parameters

<PORT-NUM>

TCP port number. Range: 1024 to 65535. Default: 6640.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring port **1055** as the manager port:

```
switch(config)# hsc
switch(config-hsc)# manager port 1055
```

Setting the manager port to the default value:

```
switch(config)# hsc
switch(config-hsc)# disable
switch(config-hsc)# no manager port
```

Accessing Aruba Support

Aruba Support Services	https://www.arubanetworks.com/support-services/
Aruba Support Portal	https://asp.arubanetworks.com/
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
Software licensing	https://lms.arubanetworks.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/
Aruba software and documentation	https://asp.arubanetworks.com/downloads

Accessing updates

To download product updates:

Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>



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Warranty information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

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